ELECTROCHEMISTRY

@ competishun YT lectures Notes.

→ It deals with interconversion of chemical action into electrical energy and vice-versa.
These interactions occur at boundaries between electrolyte and electrode dipping in it.

All these interconversions occurs in a device called electrochemical cell.

Types of Electrochemical cell

→ Galvanic cell :

- which converts the spontaneous chemical action (ΔG = -ve) into electrical energy eg. Daniel cell, dry cell, lead storage battery. etc

- Reactions involved are Redox rkn.

→ Electrolytic cell :
Electrolytic decomposition (Electrolysis) is done (ΔG = +ve) (non-spontaneous process) by providing electrical energy from external source (battery)

(OR)

A non-spontaneous rkn. is made to move or proceed forward by supplying electrical energy from an external source (Battery)

* Applications :

Electroplating, Refining of metals etc.

Electrochemistry → Galvanic cell.
Electrochemistry → Electrolytic cell.
Electrolytic conduction.

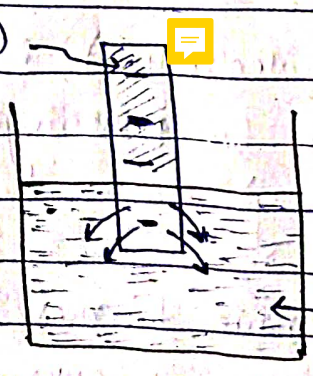
(*) Galvanic Cell :

Electrode Potential :
When a metal rod is dipped into its own ion solution, there occurs either oxidation or Reduction due to which a potential difference exist between electrode (metal Rod) and solution on interface (on surface of metal) called electrode potential.

OR

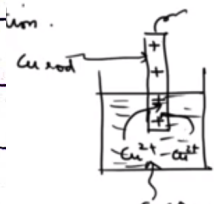
Electrode potential is a tendency to lose or to gain electron when an electrode is dipped into its own ion solution.

eg.



$Zn(s) \rightarrow Zn^{2+} + 2e^-$

($E_{oxidation}$)



E_{oxid} $Cu^{2+} + 2e^- \rightarrow Cu$

Imp

→ Absolute value of electrode potential can't not be determined. Rather relative electrode potential of various half cells are measured w.r.t a reference electrode, called Standard hydrogen electrode (S.H.E) whose Standard electrode potential at all temp. is assigned zero.

→ Electrode potential (and also cell potential or cell EMF (electromotive force) is an intensive property.

⇒ Effeciency of a galvanic cell

JEE-MAIN

$$\eta = \frac{|\Delta G|}{|\Delta H|} \times 100$$

$$= \frac{\text{Electrical work}}{\text{Heat released in rxn.}} \times 100$$

Electrical work obtained from a Galvanic cell = $\Delta G = -q \times V$

$= -n \times F \times E$

(or) $\boxed{\Delta G = -nFE}$

electrical work (or non PV work)

where F = Faraday's constant

⇒ 1F = 96,500 coulomb

= Charge present on 1 mol of electrons.

n = no. of moles of e^-

E = Electrode potential or cell Potential.

ΔG = Change in Gibb's energy for half cell rxn. or cell rxn.

⇒ Half cell potentials are tabulated usually at 25°C in a series called Electrochemical Series (ECS)

⇒ Electrode and cell potential are intensive properties,

$$\Delta G = -nF E_{cell}$$

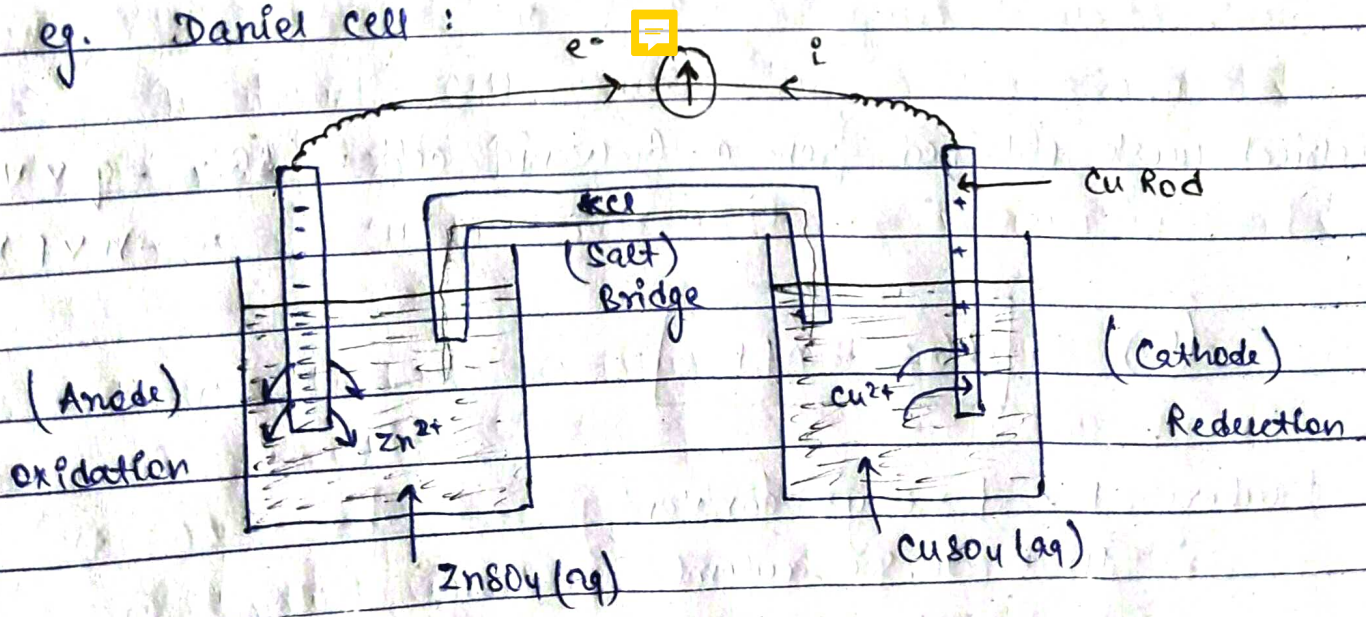
↑ Electrical work

$$\Rightarrow E_{cell} = -\frac{\Delta G}{nF}$$

↑ intensive property. ← extensive property

(*) working of Galvanic cell :

eg. Daniel cell :



$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$ ($E_{oxd.}$)

$Cu^{2+} + 2e^- \longrightarrow Cu(s)$ ($E_{Red.}$)

↓ electrons of Cu rod

$Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$, $\Delta G_1 = -2 \times F \times E_{oxidation}$

$Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$, $\Delta G_2 = -2 \times F \times E_{Reduction}$

$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$, $\Delta G_3 = \Delta G_1 + \Delta G_2$

cell reaction

$-2F \times E_{cell} = -2 \times F \times E_{oxid.} - 2 \times F \times E_{Red.}$

$\Rightarrow$ $\boxed{E_{cell} = E_{oxid} + E_{Red.}}$

→ for spontaneous working of Galvanic cell, E_{cell} should be positive and so that ΔG_{cell} is -ve.

$\Rightarrow (\Delta G)_{cell} = -nF \times E_{cell}$

-ve : Electrical work

+ve

Note : (1) Electrode potential or cell potential are intensive properties and do not depend on

- Amount of solution.
- Thickness of metal Rod / plate / wire etc.
- Does not depend on Stoichiometry variation of a given cell rxn.

$A^{2+} + 2e^- \longrightarrow A(s)$, $E = x$ volt, $\Delta G = z$ KJ/mol

$2A^{2+} + 4e^- \longrightarrow 2A(s)$, $E = x$ volt, $\Delta G = 2y$ KJ/mol

$\Delta G = -nF \times E$ → remains same.

↑ doubled ↑ doubled

⇒ E_{cell} or E (electrode potential) depends on

gm
- → Temperature.
- → conc. of solution or partial pressure of gas involved
- → Nature of electrode used.

(2) Description of Galvanic cell :

		Anode	Cathode
(1)	Reaction	Oxidation	Reduction
(2)	Sign	-ve	+ve
(3)	Direction of e^- flow	out of cell	Into cell

⇒ In daniel cell, with time, mass of Zn Rod ↓, Conc. of Zn^{2+} ↑, mass of Cu Rod increases, and conc. of Cu^{2+} ions ↓.

(3) Under Standard conditions, cell potential is represented as E°_{cell}

Standard Conditions,
Pressure of gases involved : 1 bar & assumed ideal
Conc. of Species : 1 Molar.

Solids & liquids : Pure and at 1 bar pressure.

→ Temp. is not the part of standard conditions. However, the potential state (SRP) is tabulated at 25°C (ambient temp.)

↑ Standard Reduction Potential or Standard Electrode Potential.

IUPAC Representation of Cell

eg. Daniel cell

$Zn(s) / ZnSO_4(aq) \,||\, CuSO_4(aq) / Cu$

$Zn(s) / ZnSO_4(aq)$ — Anode Part (oxidation Half)

$CuSO_4(aq) / Cu$ — Cathode Part (or Reduction Half)

→ Represent anode part to the left side and cathode part at the right side. Both parts are separated by a double vertical line representing presence of salt Bridge.

→ Electron Carrier (eg. Zn & Cu Rods in Daniel cell) must be represented at extreme left and extreme Right

→ Single vertical line (|) represents the interface between two different phases in contact.
However, comma (,) is used when two species in same phase are in contact.

e.g $Pt / Fe^{2+}_{(aq)}, Fe^{3+}_{(aq)} \,||\, Cu^{2+}_{(aq)} / Cu_{(s)}$

$Pt / Fe^{2+}_{(aq)}, Fe^{3+}_{(aq)}$ — Anode

$Cu^{2+}_{(aq)} / Cu_{(s)}$ — Cathode

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Soln (a) $Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$

$MnO_4^- + 5e^- + 8H^+ \longrightarrow Mn^{2+}(aq) + 4H_2O$

$5Fe^{2+}(aq) + MnO_4^- + 8H^+ \rightarrow 5Fe^{3+}(aq) + Mn^{2+}(aq) + 4H_2O$

Cell rkn.

(b) $2Cl^-(aq) \longrightarrow Cl_2(g) + 2e^-$

$Ce^{4+} + e^- \longrightarrow Ce^{3+}$ $\times$ (2)

$2Ce^{4+}(aq) + 2Cl^-(aq) \longrightarrow Cl_2(g) + 2Ce^{3+}(aq)$

Note: Electrodes (Rod / wire / Plate) may be of two types.

(i) Inert Electrode: Pt, graphite (acts only as e^- carrier between two halves of cell) and do not involve in cell rkn.

(ii) Active Electrodes:
which take part in cell rkn and also act as e^- carrier.

eg. Zn, Cu rods in Daniel cell.

$Cu^{2+}(aq) + Zn(s) \longrightarrow Cu(s) + Zn^{2+}(aq)$

* Salt Bridge :

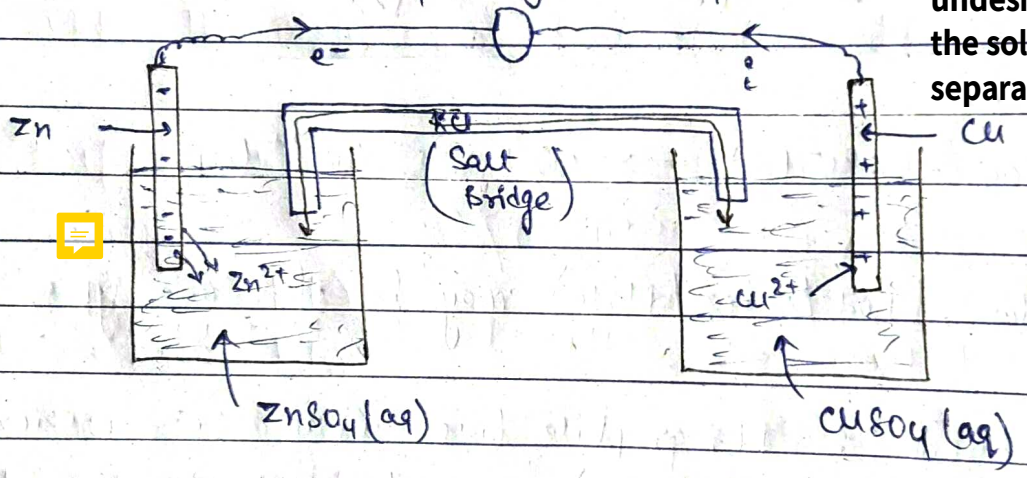
It is an inverted U shaped tube containing an electrolyte filled with highly viscous (gel like) solution with agar agar or dride.

* Function of Salt Bridge -

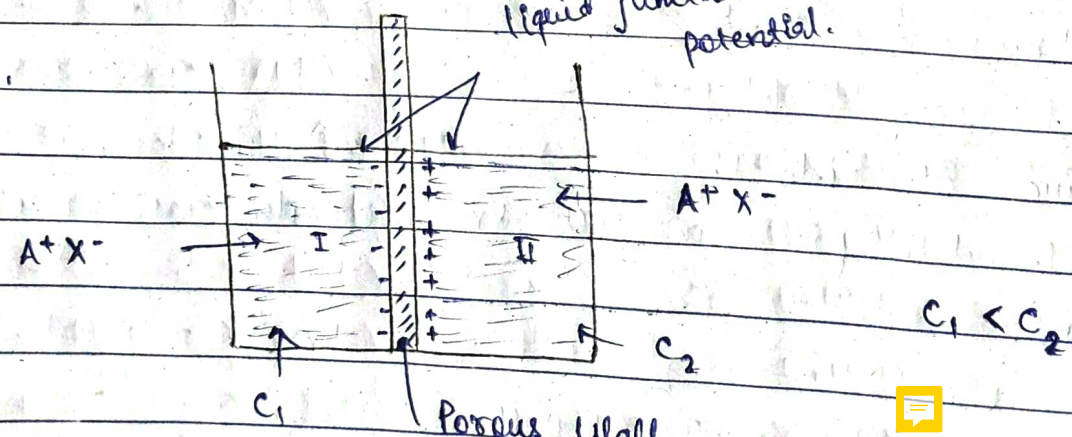
(1) To maintain anode half and cathode half electrically neutral.

(2) To minimize liquid junction potential

undesirable potential developed in the solution due to charge separation



eg.



$C_1 < C_2$

If mobility of $X^- > A^+$

* Ions of Salt Bridge Electrolyte diffuse through gel like solution and minimize liquid junction.

* Salt bridge does not allow the bulk flow of components of anode to cathode and vice verse. However a minor diffusion of components of anode or cathode can occur through Salt Bridge.

* Salt Bridge is not needed for Galvanic cell in which a common electrolyte of anode half and cathode half is present.

eg. Concentration Cell,

$$Pt / H_2 (P_1) / HCl(aq) / H_2 (P_2) / Pt$$

(No Salt bridge needed in such case)

V. Imp

* Suitable electrolyte of Salt Bridge should have equal or nearly equal ionic mobility of their ions. eg. KCl is a suitable electrolyte for such purpose.

* Electrolyte of Salt bridge should be inert towards the components of anode / cathode parts (i.e No ppt^n, No complexation rkn.)

Imp

KCl can not be used if Anode/cathode parts contains ion like Pb^{2+}, Ag^+, Tl^+, Hg^{2+} etc
(otherwise ppt. will occur)

→ Then KNO_3, NH_4NO_3 etc may be used in such case.

(*) Electrochemical Series (ECS)
(Also known as Activity Series)

→ Standard electrode potential (or Standard Reduction Potential) of various half cells are measured at 25°C w.r.t S.H.E

These electrode potential are arranged in increasing order of SRP, the series formed is called ECS.

Half cell rkn.	SRP (in Volt)
$Li^+ + e^- \rightarrow Li(s)$	− 3.05 V
$K^+ + e^- \rightarrow K(s)$	− 2.94 V
$Ca^{2+} + 2e^- \rightarrow Ca(s)$	− 2.87 V
$Na^+ + e^- \rightarrow Na(s)$	− 2.71 V
$Mg^{2+} + 2e^- \rightarrow Mg(s)$	− 2.36 V
$Al^{3+} + 3e^- \rightarrow Al(s)$	− 1.66 V
$2H_2O + 2e^- \rightarrow H_2\uparrow + 2OH^-$	− 0.83 V
$Zn^{2+} + 2e^- \rightarrow Zn(s)$	− 0.76 V
$Fe^{2+} + 2e^- \rightarrow Fe(s)$	− 0.44 V
$Sn^{2+} + 2e^- \rightarrow Sn(s)$	− 0.14 V
$Pb^{2+} + 2e^- \rightarrow Pb(s)$	− 0.13 V
$2D^+ + 2e^- \rightarrow D_2(g)$	− 0.01 V
$2H^+ + 2e^- \rightarrow H_2(g)$	0.00 V ← Reference Electrode (SRP = 0)

$Fe^{3+} + 3e^- \rightarrow Fe(s)$ − 0.04V

$Sn^{4+}(aq) + 2e^- \longrightarrow Sn^{2+}(aq)$	0.15 V
$AgCl(s) + e^- \longrightarrow Ag(s) + Cl^-(aq)$	0.22 V
$Cu^{2+} + 2e^- \longrightarrow Cu(s)$	0.34 V
$Fe^{3+}_{(aq)} + e^- \longrightarrow Fe^{2+}(aq)$	0.77 V
$Ag^+ + e^- \longrightarrow Ag(s)$	0.79 V
$Hg^{2+} + 2e^- \longrightarrow Hg(l)$	0.86 V
$Au^{3+} + 3e^- \longrightarrow Au(s)$	1.4 V
$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$	1.51 V
$F_2 + 2e^- \longrightarrow 2F^-(aq)$	2.87 V

* Applications of ECS :

(1) Species (particularly metals) having highly -ve SRP, are powerful R.A

eg. Li, Na, Al, Zn, Fe, etc.

→ Species with high (+ve) SRP are powerful Oxidising Agent (OA)

eg. F_2, MnO_4^-, Cu^{2+}, Ag^+, Fe^{3+}, Cl_2, Au^{3+} etc.

(2) More active metals can displace less active metals from their Salt solutions.

eg. $CuSO_4(aq) + Zn(s) \longrightarrow Cu(s) + ZnSO_4(aq)$

$E^\circ_{cell} = 0.34 + (0.76V)$

$= 1.1V\ (+ve)$

$\Delta G^\circ = -2 \times F \times 1.1 = -ve$ (Spontaneous)

$\Rightarrow CuSO_{4(aq)} + Cu(s) \longrightarrow$ No rxn.

$\Rightarrow E^\circ_{cell} = -1.1V$

$\Delta G^\circ = +ve$

(3) Highly active metals can displace H_2 gas from water even at room temp.

⇒ Na, K, Ca etc $\xrightarrow{+H_2O(l)}$ $H_2\uparrow$

Fe, Sn, Ni etc $\xrightarrow{+\text{steam (or } H_2O(g))}$ $H_2\uparrow$

⇒ Metals (less active) placed below 'H' in ECS do not displace H_2 from water or steam

Cu, Ag, Au, Pt $\xrightarrow{+H_2O(l) \text{ or steam}}$ No(H_2) gas evolution

(4) Metals placed above 'H' in ECS can displace H_2 gas from dilute acids (like HCl)

Na, Ca, Al, Fe, Zn, Sn $\xrightarrow{+HCl(aq)}$ $H_2\uparrow$

⇒ Cu / Ag / Au / Pt / Hg $\xrightarrow{+HCl(g)}$ No rxn.

(5) Oxides of active metals are thermally stable.

Na_2O, Al_2O_3, CaO, MgO, ZnO

(thermally stable.)

IIT

⇒ But oxides of less active metals are thermally less stable.

eg. thermally unstable metal oxides

$Ag_2O(s) \xrightarrow{\Delta} 2Ag(s) + \frac{1}{2}O_2\uparrow$

$HgO(s) \xrightarrow{\Delta} Hg(l) + \frac{1}{2}O_2\uparrow$

Q Given data, $E^\circ_{Zn^{2+}/Zn} = -0.76\,V$

$E^\circ_{Fe^{2+}/Fe} = -0.44\,V$

$E^\circ_{Cu^{2+}/Cu} = 0.34$ Volt

(i) How many cells can be formed with these combinations and find their E°cell (or cell EMF)

(ii) find maximum E°_{cell} and minimum E°_{cell}

Soln: ⇒ Remember : Make cathode whose SRP or RP is high.

$Zn^{2+}/Zn = -0.76\,V$

$Cu^{2+}/Cu = -0.34\,V$

$Fe^{2+}/Fe = -0.44\,V$

(i) Cu & Zn

$E^\circ cell = E^\circ oxid + E^\circ Red.$

$= 0.76 + 0.34\,V$

$= 1.1\,V$ (Maximum E°cell)

Cu & Fe

$E^\circ cell = 0.44 + 0.34$

$= 0.78\,V$

Zn & Fe

$E^\circ cell = 0.76 + (-0.44)$

$= 0.32\,V$

(Minimum E°cell)

Imp

Q. Given,

$E^\circ_{Cu^{2+}, Cu^+/Pt} = 0.15\ V$

$E^\circ_{Cu^+/Cu} = 0.52\ V$

Find E° of half cell rkn.

$Cu^{2+} + 2e^- \rightarrow Cu.$

Soln :

$Cu^{2+} + e^- \longrightarrow Cu^+ \quad , 0.15\ V \qquad \Delta G_1^\circ = -1 \times F \times 0.15\ V$

$Cu^+ + e^- \longrightarrow Cu \quad , 0.52\ V \qquad \Delta G_2^\circ = -1 \times F \times 0.52\ V$

$Cu^{2+} + 2e^- \longrightarrow Cu \quad , \Delta G_3^\circ = -2 \times F \times E^\circ$

$\Delta G_3^\circ = \Delta G_1^\circ + \Delta G_2^\circ$

$-2 \times F \times E^\circ = -1 \times F \times 0.15\ V - 1 \times F \times 0.52$

$$E^\circ = \frac{0.67}{2} = 0.335 \text{ Volt}$$

Ans.

∋ Electrode and cell potential are intensive properties.

V. Imp

NERNST's EQUATION :

(Variation of Electrode or cell potential with concentration, partial pressure or temperature)

eg. cell rkn. $aA + bB \longrightarrow cC + dD$

From thermodynamics,

$$\Delta G_r = \Delta G_r^\circ + 2.303RT \log_{10} Q_{pc}$$

$$-nF \times E_{cell} = -nF \times E^\circ_{cell} + 2.303\,RT \log_{10} Q$$

$$\Rightarrow \boxed{E_{cell} = E^\circ_{cell} - \frac{2.303RT}{nF} \log_{10} Q}$$ Nernst's Eqn.

At 25°C, $\frac{2.303RT}{F} = 0.0591$

$$\boxed{E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log_{10} Q}$$ Nernst eqn at 25°C.

eg. $Cu^{2+} + Zn(s) \longrightarrow Cu(s) + Zn^{+}(aq)$, $Q = \frac{[Zn^{2+}]}{[Cu^{2+}]}$

eg. $Cu^{2+}(aq) + H_2 \longrightarrow 2H^{+}(aq) + Cu(s)$, $Q_{pc} = \frac{[H^+]^2 \times 1}{[Cu^{2+}] \times p_{H2}}$

* Applications of Nernst equation :

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \log_{10} Q$$

→ Applicable for cell as well as Half cell Rkn.

(1) To determine non-standard cell potential or electrode potential

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \log_{10} Q$$

E_{cell} ↑ Non-standard cell Potential

E°_{cell} ↑ Standard cell potential.

(2) To determine equilibrium constant

$$\Delta G = \Delta G^{\circ} + 2.303RT \log_{10} Q$$

OR

$$E_{cell} = E^{\circ}_{cell} - \frac{0.0591}{n} \log_{10} Q$$

At equilibrium, $\Delta G = 0$ | No electrical work is obtained once eqm. is established)

Imp

$$E_{cell} = 0 \quad (\text{At eqm.})$$

$$Q = K_{eq}$$

$$0 = E^{\circ}_{cell} - \frac{0.0591}{n} \log_{10} K_{eq}$$

From here, we can calculate K_{eq}.

* Problems on Nernst's equation :-

Q. Find half cell potential of the following

(a) $Mg/Mg^{2+}(aq)$ $(2\times10^{-3}M)$, Given $E^\circ_{Mg^{2+}/Mg} = -2.36\,V$

(b) $Cl^-(aq)/Cl_2(g)/Pt$ $(10^{-2}M)$ $(1\,Bar)$, Given $E^\circ_{Cl^-/Cl_2/Pt} = +1.35\,V$

Soln : (a) $Mg \longrightarrow Mg^{2+} + 2e^-$, $E^\circ = +2.36\,V$

$$E = 2.36 - \frac{0.06}{2}\log(2\times10^{-3})$$

$$= 2.36 - 0.03\,[-3 + 0.3]$$

$$= 2.36 + 0.03 \times 2.7$$

$$= 2.36 + 0.081$$

$$= 2.441 \text{ Volt Ans.}$$

(b) $Cl_2(g) + 2e^- \longrightarrow 2Cl^-(aq)$, 1.35 Volt

1 Bar $\qquad\qquad$ $10^{-2}M$

$$E = 1.35 - \frac{0.06}{2}\log\frac{(10^{-2})^2}{1}$$

$$= 1.35 - 0.03 \times (-4)$$

$$= 1.35 + 0.12 = 1.47 \text{ Volt}$$

Q. Find cell EMF

$$Zn \,|\, ZnSO_4 \,(4\times10^{-3}M) \;\|\; CuSO_4\,(aq)\,(0.1M) \,|\, Cu$$

$E^\circ_{Cu^{2+}/Cu} = 0.34$

$E^\circ_{Zn^{2+}/Zn} = -0.76\,V$

Soln:

$$Zn \longrightarrow Zn^{2+} + 2e^-, \quad E^\circ = 0.76\,V$$

$$Cu^{2+} + 2e^- \longrightarrow Cu, \quad E^\circ = 0.34\,V$$

Cell Rkn. $Cu^{2+} + Zn \longrightarrow Zn^{2+} + Cu$, $E^\circ_{cell} = 1.14$ Volt

(Cu^{2+}: 0.1M; Zn^{2+}: $4\times10^{-3}M$)

$$E = 1.1 - \frac{0.06}{2}\log\frac{4\times10^{-3}}{0.1}$$

$$= 1.1 - 0.03\log(4\times10^{-2}) = 1.1 - 0.03\times(-1.4)$$

$$= 1.1 + 0.042$$

$$= 1.142 \text{ Volt}$$

Q. Find E_{cell}

$$Pt \,/\, H_2(g) \,/\, CH_3COOH\,(aq)\,(0.1M) \;\|\; MnO_4^-\ (0.1M),\ Mn^{2+}\ (0.1M),\ H^+\ (10^{-2}M) \,/\, Pt$$

$E^\circ_{MnO_4^-, Mn^{2+}, H^+/Pt} = 1.5\,V$

$Ka(CH_3COOH) = 10^{-5}$

Soln:

$$H_2(g) \longrightarrow 2H^+(aq) + 2e^-(aq),\ E^\circ = 0 \quad (\times 5)$$

$$MnO_4^- + 5e^- + 8H^+ \longrightarrow Mn^{2+} + 4H_2O(l) \quad (\times 2)$$

$0.1M$ $(10^{-2}M)$ $0.1M$

$$2MnO_4^- + 16H^+ + 5H_2(g) \longrightarrow 10H^+ + 2Mn^{2+} + 8H_2O$$

$0.1M$ $(10^{-2}M)$ (c) $0.1M$

$\downarrow 10^{-3}M$

$$E = (0 + 1.5) - \frac{0.06}{10} \log \frac{(10^{-3})^{10} \times (0.1)^2}{(0.1)^2 \times (10^{-2})^{16} \times (1)^5}$$

$$E = 1.5 - 0.006 \log 10^2$$

$$= 1.5 - 0.006 \times 2$$

$$= 1.5 - 0.012$$

$$= 1.488\ V$$

$$[H^+] = \sqrt{K_a \times C}$$

$$= \sqrt{10^{-5} \times 0.1}$$

$$= 10^{-3}M$$

from CH_3COOH

Find oxidation Potential of Hydrogen gas electrode at $pH = 7$

$$H_2(g) \longrightarrow \underset{(10^{-7}M)}{2H^+(aq)} + 2e^-$$

$$E_{oxid} = 0 - \frac{0.06}{2} \log 10^{-14}$$

$$= -0.03(-14)$$

$$= 0.42\ Volt.$$

Q. Find $\log_{10} K_{eq}$ for the following cell rxn. at eqm.

$$Sn(s) + Sn^{4+}(aq) \rightleftharpoons 2Sn^{2+}(aq)$$

Given - $E^\circ_{Sn^{4+}/Sn^{2+}/Pt} = 0.15\ V$

$E^\circ_{Sn^{2+}/Sn} = -0.14\ V$

Soln : $E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log Q$

At eqm, $E_{cell} = 0$, $Q = K_{eq}$

$$0 = (0.14 + 0.15) - \frac{0.06}{2} \log_{10} K_{eq}$$

$$\Rightarrow 0.03 \log K_{eq} = 0.29$$

$$\log K_{eq} = 29/3 = 9.67 \text{ Ans.}$$

gmd.

Q. Find E° of $[Ag(NH_3)_2]^+ + e^- \longrightarrow Ag(s) + 2NH_3(aq)$ — (2)

Given $E^\circ_{Ag^+/Ag} = 0.8\ V$

$$[Ag(NH_3)_2]^+ \rightleftharpoons Ag^+(aq) + 2NH_3(aq)$$

$$K_{eq} = 10^{-14}$$

Soln: $Ag^+ + e^- \longrightarrow Ag$, 0.8 V

$Ag \longrightarrow Ag^+ + e^-$, −0.8 V — (1)

(1) + (2)

$$0 = (-0.8 + x) - \frac{0.06}{1} \log 10^{-14}$$

$$\Rightarrow x = 0.8 + 0.06 \log 10^{-14}$$

$$= 0.8 - 0.06 \times 14$$

$$= 0.8 - 0.84$$

$$= 0.04 \text{ Volt}$$ Ans

* Concentration cell :

A galvanic cell in which cathode and anode halves are made up of same material, current flows in outer circuit due to difference in conc. of solution or partial pressure of gas involves. difference of active masses b/w the two halves

* E°_{cell} is always zero for a conc. cell.

eg. of concentration cell,

$Zn \,|\, ZnSO_4\,(aq) \,\|\, ZnSO_4\,(aq) \,|\, Zn$

(C_1) (C_2)

$C_1 \neq C_2$

concentration cell

eg. $Pt \,/\, H_2 \,/\, HCl\,(aq) \,\|\, HCl\,(aq) \,/\, H_2\,(g) \,/\, Pt$

P_1 (C_1) (C_2) P_2

anode cathode

$$\underset{(P_1)}{H_2} \longrightarrow \underset{(C_1)}{2H^+} + 2e^- \quad , \quad E^\circ_{Oxd} = 0$$

$$\underset{(C_2)}{2H^+} + 2e^- \longrightarrow \underset{P_2}{H_2(g)} \quad , \quad E^\circ_{Red.} = 0$$

Cell Reaction :

$$\underset{P_1}{H_2(g)} + \underset{(C_2)}{2H^+(aq)} \longrightarrow \underset{(C_1)}{2H^+(aq)} + \underset{P_2}{H_2(g)} \quad , \quad E^\circ_{cell} = 0$$

$$E_{cell} = 0 - \frac{0.0591}{2} \log \left(\frac{C_1^2 \times P_2}{C_2^2 \times P_1} \right)$$

For spontaneous working of cell, E_{cell} should be positive

$$\frac{C_1^2 \times P_2}{C_2^2 \times P_1} < 1$$

$$\Rightarrow \left[C_1^2 \times P_2 < C_2^2 \times P_1 \right]$$

Q. Find cell EMF of following conc. cell,

$$Pt \,|\, \underset{10^{-4}M}{Cl_2 / KCl(aq)} \,\|\, \underset{(0.1M)}{HCl(aq)} \,|\, \underset{(2atm)}{Cl_2} \,|\, Pt$$

Soln

$$\underset{10^{-4}}{2Cl^-} \longrightarrow \underset{1atm}{Cl_2} + 2e^- \quad , \quad E^\circ_{Oxid} = x$$

$$\underset{2atm}{Cl_2} + 2e^- \longrightarrow \underset{0.1M}{2Cl^-} \quad , \quad E^\circ_{Red} = -x$$

$$\underset{2atm}{Cl_2} + \underset{10^{-4}M}{2Cl^-} \longrightarrow \underset{1atm}{Cl_2} + \underset{0.1M}{2Cl^-}$$

$$E_{cell} = 0 - \frac{0.06}{2} \log \frac{1 \times 10^{-2}}{2 \times 10^{-8}}$$

$$= -0.03 \log (5 \times 10^5)$$

$$= -0.03 (\log 5 + 5)$$

$$= -0.03 \times 5.7 = -0.171 \text{ volt}$$

(Non - Spontaneous)

Q. In which case, E_{cell} is maximum -

(A) $Pt / H_2 / HCl$ (0.1M) $\|$ S.H.E

(B) $Pt / H_2 / H_2SO_4$ (0.1M) $\|$ S.H.E

(3) ~~$Pt / H_2 / CH_3COOH$ (0.1M) $\|$ S.H.E~~

(D) $Pt / H_2 / H_3PO_4$ (0.1M) $\|$ S.H.E

Soln $\frac{1}{2} H_2 \longrightarrow H^+ + e^-$

$$E_{oxid} = 0 - \frac{0.0591}{1} \log [H^+], \quad E_{Red} = E^\circ_{Red} = 0$$

$$E_{cell} = E_{oxid} + 0$$

$$= E_{oxid}$$

$[H^+] \downarrow$, $E_{oxid} \uparrow$

(In General)

$$E_{cell} = E^\circ_{cell} - \frac{0.0591}{n} \log \frac{[Prod]}{[React]}$$

$E_{cell} \uparrow$ If $[Prod] \downarrow$

$[Reactant] \uparrow$

$C > D > A > B$

$\longrightarrow$

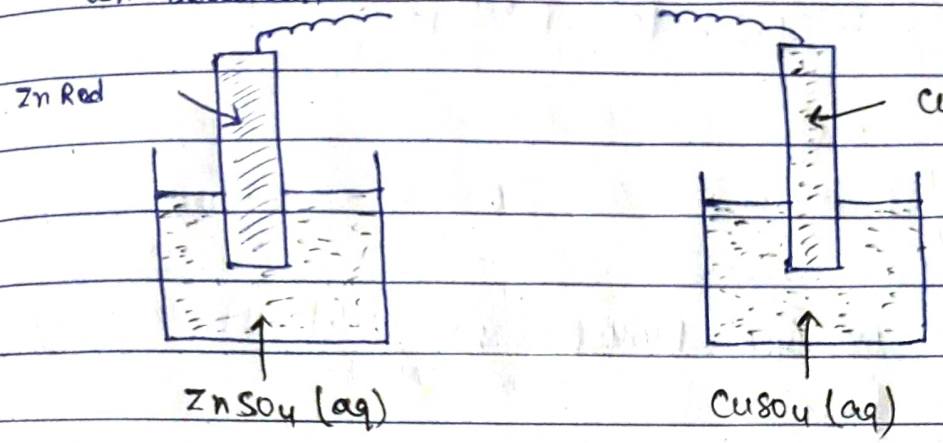
decreasing cell EMF.

$K_a(CH_3COOH) < K_a(H_3PO_4)$

* Various Types of Half Cells or Electrodes :-

① Metal - Metal ion Half cell :-

→ It consists of a metal rod/plate dipped into its own ion solution.

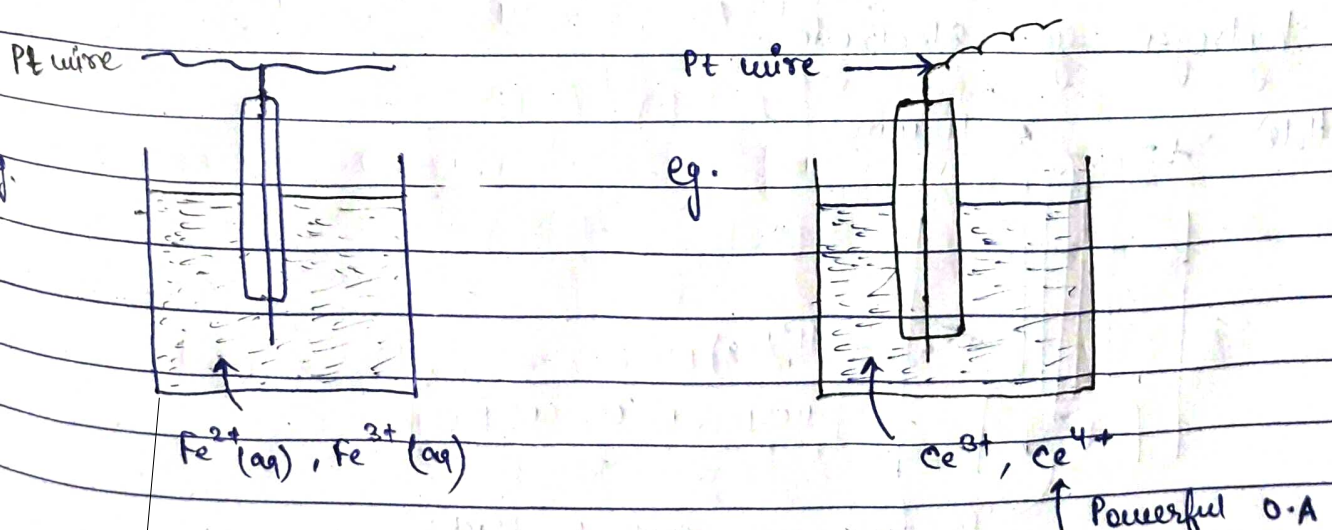


$Zn \longrightarrow Zn^{2+} + 2e^-$, $Zn / Zn^{2+}(aq)$ [Anode]

$Zn^{2+}(aq) + 2e^- \longrightarrow Zn(s)$, Zn^{2+}/Zn [cathode]

② Redox Half cell :-

In such half cell, an element (metal/non-metal) is present in aq. solution in two different oxidation ~~se~~ states. Pt wire is used as electron carrier.



If it acts as Anode,

$Fe^{2+}(aq) \longrightarrow Fe^{3+}(aq) + e^-$, $Pt \mid Fe^{2+}, Fe^{3+}$

If it acts as Cathode,

$Fe^{3+}(aq) + e^- \longrightarrow Fe^{2+}(aq)$, $Fe^{3+}, Fe^{2+} \mid Pt$

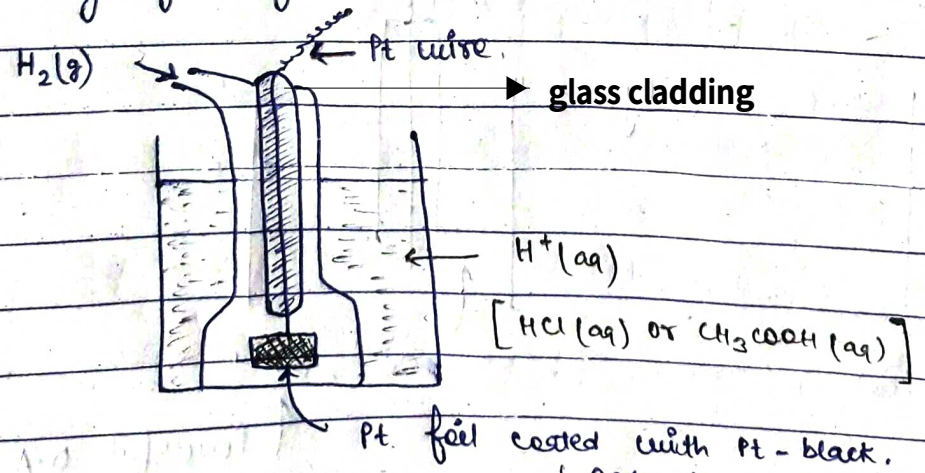
eg. $MnO_4^-, Mn^{2+}, H^+ \mid Pt$

(3) Gas Half cell or Gas Electrode :

→ It consists of a gas adsorbed on an inert metal (like Pt) coated with finely powdered Pt (Occlusion ⇒ adsorption of a gas on powdered Pt black)
Powdered Pt black coating has high surface area and hence greater adsorption of gas.
This set-up is dipped into an aq. solution containing ion of gaseous species.

Reduction or oxidation occurs on Pt foils.

eg. hydrogen gas Electrode —



If it acts as anode,

$H_2(g) \longrightarrow 2H^+(aq) + 2e^-$, $Pt/H_2(g)/HCl(aq)$, $E°_{SOP} = 0$
(P) (C)

If it acts as cathode,

$2H^+(aq) + 2e^- \longrightarrow H_2(g)$, $E°_{SRP} = 0$

$HCl(aq)/H_2(g)/Pt$

eg. Cl_2 gas electrode,

$Cl_2(g) + Cl^-(aq)$
$H_2(g) + H^+(aq)$
$O_2(g) + OH^-(aq)$
$Br_2(g) + Br^-(aq)$

If anode,

$Pt / Cl_2(g) / Cl^-(aq)$

If acts as cathode,

$\underset{(c)}{Cl^-(aq)} / \underset{(p)}{Cl_2} / Pt$

eg. $Pt / Cl_2 / Cl^-(aq) \;||\; H^+ / H_2 / Pt$

⇓

Cell Rkn.

$2Cl^- \longrightarrow Cl_{2(g)} + 2e^-$

$2H^+ + 2e^- \longrightarrow H_2$

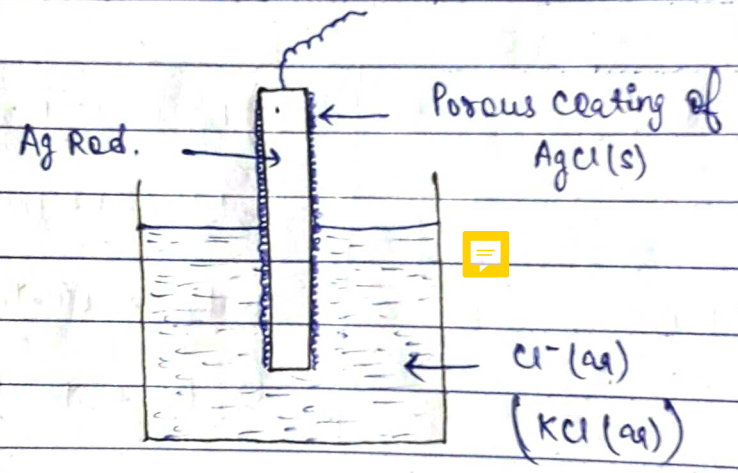
Cell Rkn. : $2Cl^-(aq) + 2H^+(aq) \longrightarrow Cl_2(g) + H_2(g)$

(4) V-imp Metal - metal insoluble Salt - anion Half cell -

$Ag - AgCl - Cl^-(aq)$

$Pb - PbSO_4 - SO_4^{2-}(aq)$

⇒ It consists of a metal Rod coated with metal insoluble salt and dipped into an aq. solution containing anion of insoluble salt.



If it acts as Cathode (Reduction Half)

$AgCl(s) + e^- \longrightarrow Ag(s) + Cl^-(aq)$

IUPAC

$Cl^-(aq) \,|\, AgCl(s) \,|\, Ag(s)$

If it acts as Anode (oxidation Half)

$Ag(s) + Cl^-(aq) \longrightarrow AgCl(s) + e^-$

IUPAC

$Ag \,|\, AgCl(s) \,|\, Cl^-(aq)$

* Determination of E° of Metal- Metal insoluble salt - anion Half cell :-

Eg: (let us take $Cl^-/AgCl/Ag$)

↳ acting as Cathode

Cell rkn.: $AgCl(s) + e^- \longrightarrow Ag(s) + Cl^-(aq)$, $E^\circ = ?$

$Ag^+(aq) + e^- \longrightarrow Ag(s)$, $E^\circ_{Ag^+/Ag}$, $\Delta G_1^\circ = -n \times F \times E^\circ_{Ag^+/Ag}$

$AgCl(s) \longrightarrow Ag^+(aq) + Cl^-(aq)$, $Ksp(AgCl)$,

$\Delta G_2^\circ = -2.303RT \log_{10} Ksp$

(+)

$AgCl(s) + e^- \longrightarrow Ag(s) + Cl^-(aq)$, $\Delta G_3^\circ = \Delta G_1^\circ + \Delta G_2^\circ$

$-n \times F \times E^\circ_{Cl^-/AgCl/Ag} = -nFE^\circ_{Ag^+/Ag} - 2.303RT \log Ksp$

$\Rightarrow E^\circ_{Cl^-/AgCl/Ag} = E^\circ_{Ag^+/Ag} + \frac{2.303RT}{nF} \log Ksp$

↓

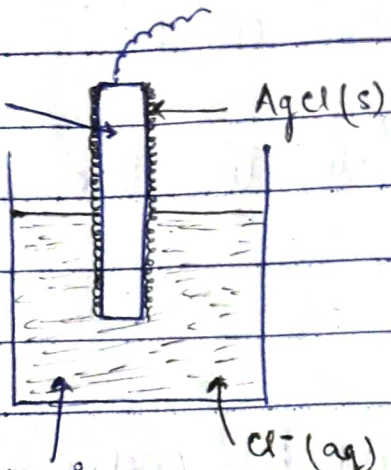
1M (as in standard conditions)

$\Rightarrow E^\circ_{Cl^-/AgCl/Ag} = E^\circ_{Ag^+/Ag} + \frac{0.0591}{n} \log_{10} Ksp\,(AgCl)$

↑

Standard half cell Potential of Metal- Metal insoluble Salt- anion half cell.

* Determination of non-standard potential of Metal - Metal insoluble salt - anion half cell :-



Solution is Saturated w.r.t AgCl

$[Ag^+] \times [Cl^-] = K_{sp}(AgCl)$

Half Cell Rkn.

$AgCl(s) + e^- \longrightarrow Ag(s) + Cl^-(aq)$

Applying Nernst eqn,

$E_{Cl^-/AgCl/Ag}$

$= E^\circ_{Cl^-/AgCl/Ag} - \frac{0.0591}{n} \log [Cl^-]$

$= E^\circ_{Ag^+/Ag} + \frac{0.0591}{n} \log K_{sp} - \frac{0.0591}{n} \log [Cl^-]$

$= E^\circ_{Ag^+/Ag} - \frac{0.0591}{n} \log \frac{[Cl^-]}{K_{sp}}$

$\frac{[Cl^-]}{K_{sp}(AgCl)} = \frac{1}{[Ag^+]}$

$\Rightarrow E_{Cl^-/AgCl/Ag} = E^\circ_{Ag^+/Ag} - \frac{0.0591}{n} \log \frac{1}{[Ag^+]}$

What is this?

$= E_{Ag^+/Ag}$

where $[Ag^+] = \frac{K_{sp}(AgCl)}{[Cl^-]}$

$Ag^+ + e^- \rightarrow Ag,$

$E_{Ag^+/Ag} = E^\circ_{Ag^+/Ag} - \frac{0.0591}{1} \log \frac{1}{(Ag^+)}$

reduction reduction

eg. $E_{SO_4^{2-}/PbSO_4/Pb} = E_{Pb^{2+}/Pb}$

↑ c mol/lit

where $[Pb^{2+}] = \frac{K_{sp}(PbSO_4)}{[SO_4^{2-}]}$

$= \frac{K_{sp}(PbSO_4)}{c}$

oxidation oxidation

eg. $E_{Pb/PbSO_4/SO_4^{2-}(aq)} = E_{Pb/Pb^{2+}(aq)}$

where $[Pb^{2+}] = \frac{K_{sp}(PbSO_4)}{[SO_4^{2-}]}$

Reduction Potential Reduction potential.

eg. $E_{CrO_4^{2-}(aq)/Ag_2CrO_4(s)/Ag(s)} = E_{Ag^+/Ag}$

where $[Ag^+]^2 \times [CrO_4^{2-}] = K_{sp}(Ag_2CrO_4)$

$\Rightarrow [Ag^+] = \sqrt{K_{sp}[CrO_4^{2-}]}$

eg. $E^\circ_{Cl^-/AgCl/Ag} = E_{Ag^+/Ag}$ where $[Ag^+] = \frac{K_{sp}}{1}$

↑ (1M)

$\Downarrow$

$Ag^+ + e^- \longrightarrow Ag$

$E_{Ag^+/Ag} = E^\circ_{Ag^+/Ag} - \frac{0.0591}{1} \log \frac{1}{[Ag^+]}$

Revision :

$$E_{Br^-/AgBr/Ag} = E_{Ag^+/Ag} \text{ where } [Ag^+] = \frac{K_{sp}(AgBr)}{[Br^-]}$$

$$= \frac{K_{sp}}{c}$$

$$= E^\circ_{Ag^+/Ag} - \frac{0.0591}{1} \log \frac{1}{[Ag^+]}$$

eg. $E^\circ_{Ag/Ag_2CO_3(s)} \,|\, CO_3^{2-}(aq) = E_{Ag/Ag^+(aq)}$

anode

1M

where $[Ag]^+ = \sqrt{\frac{K_{sp}(Ag_2CO_3)}{[CO_3^{2-}]}}$

$Ag \longrightarrow Ag^+ + e^-$

$= \sqrt{\frac{K_{sp}}{1}}$

Q. Find half cell potential of the following,

$$AgI(s) + e^- \longrightarrow Ag(s) + \underset{(0.5M)}{I^-(aq)}$$

Given $E^\circ_{Ag^+/Ag} = 0.8V$, $K_{sp}(AgI) = 10^{-18}$

Soln $E_{I^-/AgI/Ag} = E_{Ag^+/Ag}$, where $[Ag^+] = \frac{10^{-18}}{0.5}$

$= 2 \times 10^{-18}$ M.

$Ag^+ + e^- \longrightarrow Ag$

$$= E^\circ_{Ag^+/Ag} - \frac{0.06}{1} \log \frac{1}{[Ag^+]}$$

$$= 0.8 - 0.06 \log \frac{1}{2 \times 10^{-18}}$$

$$= 0.8 - 0.06 [18 - 0.3]$$

$$= 0.8 - 0.06 \times 17.7 = 0.8 - 1.062$$

$$= -0.262 \text{ Volt}$$

Emf

Q Find cell emf, cell rxn. of

$$Ag / AgI / KI_{(0.1M)} \,||\, KCl / AgCl / Ag_{(10^{-3}M)}$$

Given $K_{sp}(AgCl) = 10^{-10}$, $K_{sp}(AgI) = 10^{-18}$

Soln:

$$\underbrace{Ag / AgI / KI}_{0.1M} \,||\, \underbrace{KCl / AgCl / Ag}_{10^{-3}M}$$

$$\Downarrow \qquad\qquad \Downarrow$$

$$Ag / Ag^+(aq) \,||\, Ag^+(aq) / Ag$$

$$C_1 \qquad\qquad C_2$$

$$C_1 = \frac{10^{-18}}{0.1} = 10^{-17} M$$

$$C_2 = \frac{10^{-10}}{10^{-3}} = 10^{-7} M$$

$Ag \longrightarrow Ag^+(c_1) + e^-$, $E^\circ_{oxid} = x$

$Ag^+(c_2) + e^- \longrightarrow Ag$, $E^\circ_{red} = -x$

$Ag^+(c_2) \longrightarrow Ag^+(c_1)$, $E^\circ_{cell} = 0$

$$E = 0 - \frac{0.06}{1} \log \frac{c_1}{c_2} = +0.06 \log \frac{10^{-17}}{10^{-7}}$$

$$= -0.06 \log 10^{-10}$$

$$= 0.6 \text{ Volt}$$

Not a Conc. cell.

⇓

Cell Rkn: $Ag \,/\, AgI \,/\, KI \,\|\, KCl \,/\, AgCl \,/\, Ag$

$Ag + I^-(aq) \longrightarrow AgI(s) + e^-$

$AgCl + e^- \longrightarrow Ag(s) + Cl^-(aq)$

$AgCl(s) + I^-(aq) \longrightarrow AgI(s) + Cl^-(aq)$

Q. Write cell Rkn. and find cell emf -

$$Ag \,/\, AgCl \,/\, \underset{(10^{-2}\,M)}{KCl\,(aq)} \,\|\, \underset{(10^{-2}\,M)}{AgNO_3\,(aq)} \,/\, Ag$$

$K_{sp}(AgCl) = 10^{-10}$, $E^\circ_{Ag^+/Ag} = 0.8\,V$

Soln: $Ag(s) + Cl^-(aq) \longrightarrow AgCl(s) + e^-$

$Ag^+(aq) + e^- \longrightarrow Ag(s)$

$Ag^+(aq) + Cl^-(aq) \longrightarrow AgCl$ Cell Rkn.

$$Ag \,|\, AgCl \,|\, \underset{(10^{-2}M)}{KCl} \;\|\; \underset{(10^{-2}M)}{AgNO_3\,(aq)} \,|\, Ag$$

$$\Downarrow$$

$$Ag \,|\, Ag^+(c) \;\|\; \underset{(10^{-2}M)}{Ag^+} \,|\, Ag \qquad \text{where } c = \frac{10^{-10}}{10^{-2}}$$

$$c = 10^{-8}\ M$$

$$\underset{(10^{-2}M)}{Ag^+(aq)} \longrightarrow \underset{(c)}{Ag^+(aq)}, \quad E^\circ_{cell} = 0$$

$$E = 0 - \frac{0.06}{1} \log \frac{c}{10^{-2}}$$

$$= -0.06 \log \frac{10^{-8}}{10^{-2}}$$

$$= -0.06 \log 10^{-6}$$

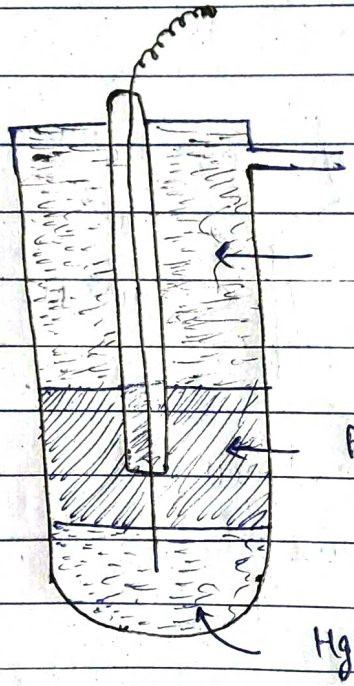
$$= 0.36 \text{ Volt.} \quad \text{Ans}$$

(5) Calomel Half Cell : (or Calomel electrode)

Calomel : $Hg_2Cl_2(s)$
(Mercurous Chloride)
• White insoluble solid

→ It is also a metal - metal insoluble Salt - anion half cell. (It is commonly used as reference electrode like S.H.E or quinhydrone electrode in potentiometric titration)

→



$[Hg_2^{2+}] \times [Cl^-]^2$
$= K_{sp}(Hg_2Cl_2)$

* All mercurous Salts involving Cl^-, Br^-, NO_3^-, I^- etc are diamagnetic compounds.

$Hg_2Cl_2(s) \rightleftharpoons Hg_2^{2+}(aq) + 2Cl^-(aq)$ [correct]

~~$Hg_2Cl_2(s) \rightleftharpoons 2Hg^+(aq) + 2Cl^-(aq)$~~ [wrong]

$Hg_2^{2+} \Rightarrow [Hg - Hg]^{2+}$
dimeric form
diamagnetic cation.

If it acts as anode (oxidation Half)

$$2Hg(l) + 2Cl^-(aq) \longrightarrow Hg_2Cl_2(s) + 2e^-$$

↳ IUPAC

$$Pt|Hg(l)|Hg_2Cl_2(s)|Cl^-(aq)$$

* If it acts as cathode,

$$Hg_2Cl_2(s) + 2e^- \longrightarrow 2Hg(l) + 2Cl^-(aq)$$

↳ IUPAC:

$$Cl^-(aq)|Hg_2Cl_2(s)|Hg(l)|Pt$$

NOTE :→ Normal or Standard Calomel Electrode

⇒ when [KCl] = 1M

→ Saturated Calomel Electrode

⇒ When KCl solution is saturated w.r.t KCl

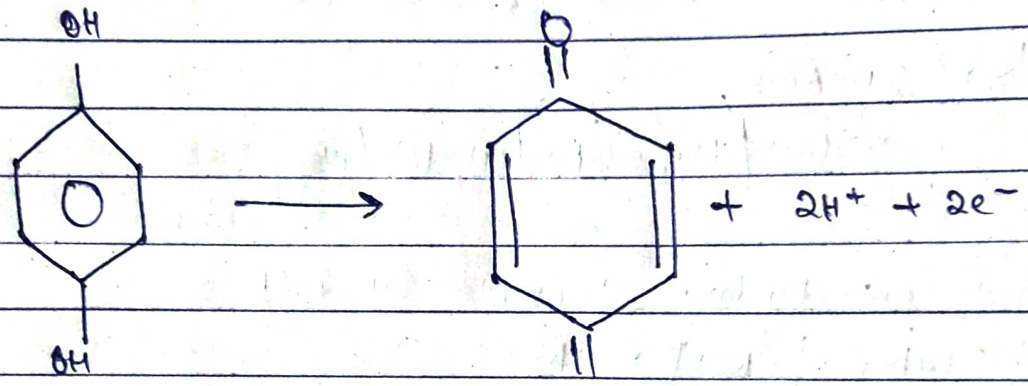
In both cases aq. KCl Sol^n is saturated w.r.t $Hg_2Cl_2(s)$

$$[Hg_2^{2+}] \times [Cl^-]^2 = K_{sp}$$

(6) Quinhydrone Electrode :

→ It is used to determine pH of a solution in the range (1 to 9)

→ A substance called quinhydrone is used to make such half cell which consists of equimolar concentration of hydroquinone and quinone in saturated solution.



Hydroquinone (H_2Q) (Quinone (Q))

$$[H_2Q] = [Q] \quad \{\text{always maintained}\}$$

$$E = E^\circ - \frac{0.0591}{2} \log \frac{[H^+]^2 \times [Q]}{[H_2Q]}$$

$$\Rightarrow E = E^\circ - \frac{0.0591}{2} \log [H^+]^2$$

$$\Rightarrow \boxed{E = E^\circ + 0.0591 \times pH}$$

Q. Calculate cell EMF when pH = 5

$Pt / H_2Q, Q, H^+ \parallel KCl(1M) \mid Hg_2Cl_2(s) \mid Hg(l) / Pt$

E°_{SRP} of Calomel electrode = 0.24 V

E°_{SRP} of ~~quinhydrone~~ = 0.74 V

Soln: $H_2Q \longrightarrow Q + 2H^+ + 2e^-$, -0.74 V

$Hg_2Cl_2(s) + 2e^- \longrightarrow 2Hg(l) + 2Cl^-(aq)$, 0.24 V

$H_2Q + Hg_2Cl_2(s) \longrightarrow Q + 2Hg(l) + 2H^+ + 2Cl^-$,

$E^\circ_{cell} = -0.74 + 0.24$

$= -0.5$ V

$$E = -0.5 - \frac{0.06}{2} \log \frac{[Q] \times [H^+]^2 \times [Cl^-]^2}{[H_2Q]}$$

$$= -0.5 - 0.03 \log (10^{-5})^2 \times (1)^2$$

$$= -0.5 + 0.3 = -0.2 \text{ Volt}$$ Ans.

* Determination of thermodynamic function ($\Delta H, \Delta G, \Delta S$) for cell reaction :

$aA + bB \longrightarrow cC + dD$, E_{cell} (cell Rkn.)

$$\boxed{\Delta G_r = -nF E_{cell}}$$

* for a substance, $dG = Vdp - SdT$

for a Chemical Rkn., $d(\Delta G) = (\Delta V)dp - (\Delta S)dT$

→ At constant pressure (or constant active masses of species involved)

$$dp = 0$$

Now, $d(\Delta G) = -(\Delta S)\, dT$

$$\Delta S_r = \frac{-d}{dT}(\Delta G)$$

$$= \frac{-d}{dT}(-nF E_{cell})$$

$$\boxed{\Delta S_r = nF \frac{dE_{cell}}{dT}}$$

where $\frac{dE_{cell}}{dT}$ = Temp. coefficient of cell emf (unit : V/K or V/°C)

$\Rightarrow \Delta G_r = \Delta H_r - T\Delta S_r$

$\Rightarrow$ $$\boxed{\Delta H_r = -nFE_{cell} + T \times nF \frac{dE_{cell}}{dT}}$$

$\Rightarrow$ Determination of ΔC_p for cell Rkn.

$$C_p = \frac{dH}{dT} \quad (\text{for a substance})$$

for a chemical reaction:

$$(\Delta C_p)_r = \frac{d}{dT}(\Delta H) \Rightarrow \text{"Kirchoff's eqn"}$$

$$= \frac{d}{dT}\left(-nFE_{cell} + nFT\frac{dE_{cell}}{dT}\right)$$

$$\Rightarrow (\Delta C_p)_r = -nF\frac{dE_{cell}}{dT} + nF\frac{dE_{cell}}{dT} + nFT\frac{d^2E_{cell}}{dT^2}$$

$$\Rightarrow \boxed{(\Delta C_p)_r = nFT\frac{d^2E_{cell}}{dT^2}}$$

Q. Reaction: $A(s) + B^{2+}(aq) \longrightarrow A^{2+}(aq) + B(s)$, $E_{cell} = 1$ volt at 25°C, $\frac{dE_{cell}}{dT} = 5 \times 10^{-5}$ V/K. Find ΔG, ΔH, ΔS.

Soln: $n = 2$, $\Delta G = -2 \times 96500 \times 1 = -193$ kJ/mol.

$$\Delta S = nF \frac{dE_{cell}}{dT} = 2 \times 96500 \times 5 \times 10^{-5}$$

$$= 9.65 \text{ J/K/mol}$$

$$\Delta H = \Delta G + T\Delta S$$

$$= -193 + 298 \times \frac{9.65}{1000} \text{ kJ/mol}$$

$$= \{-193 + 0.298 \times 9.65\} \text{ kJ/mol}$$

IIT-2001 (Subjective)

Q. The Standard Potential of the following cell is 0.23 V at 15°C and 0.21 V at 35°C.

$$Pt / H_2 / HCl(aq) \,|\, AgCl(s) / Ag(s)$$

(i) Write the cell Rxn.

(ii) Calculate ΔH°, ΔS° for the cell Rxn. by asuming that these quantities remain unchanged in the range 15°C to 35°C.

(iii) Calculate Solubility of AgCl in water at 25°C.

Given $E^\circ_{Ag^+/Ag} = 0.81$ Volt, $\frac{2.303RT}{F} = 0.059$

Sol^{n} : $E^\circ_{15^\circ C} = 0.23\ V$, $E^\circ_{35^\circ C} = 0.21\ V$

no salt bridge as common solution of HCl(aq)

$Pt / H_2 / HCl(aq) / AgCl(s) / Ag$

(i) $H_2(g) \longrightarrow 2H^+ + 2e^-$

$AgCl + e^- \longrightarrow Ag(s) + Cl^-(aq)$ $\times 2$

$H_2 + 2AgCl(s) \longrightarrow 2H^+(aq) + 2Ag(s) + 2Cl^-(aq)$ (cell rkn.)

(ii) $\Delta H_0 = ?$, $\Delta S^\circ = ?$

$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$

$-2 \times 96500 \times 0.23 = \Delta H^\circ - 288 \times \Delta S^\circ$

$-2 \times 96500 \times 0.21 = \Delta H^\circ - 308 \times \Delta S^\circ$

$\Delta S^\circ =$ ✓

$\Delta H^\circ =$ ✓

(iii) At 25°C, $E^\circ_{cell} = \dfrac{0.23 + 0.21}{2} = 0.22$ Volt

$E^\circ_{cell} = E^\circ_{Pt/H_2/H^+} + E^\circ_{Cl^-/AgCl/Ag}$

$0.22 = 0 + E^\circ_{Cl/AgCl/Ag} \Rightarrow = E_{Ag^+/Ag}$

$\Rightarrow 0.22 = E^\circ_{Ag^+/Ag} + \dfrac{0.0591}{1} \log \dfrac{1}{[Ag^+]}$

$$0.22 = 0.81 + 0.059 \log \frac{1}{\left(\frac{K_{sp}}{1}\right)}$$

$$\Rightarrow \frac{-0.59}{0.059} = \log K_{sp}$$

$$\Rightarrow K_{sp} = 10^{-10}$$

$$AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$$
$$\quad s \qquad\qquad\quad s \qquad\quad s$$

$$s^2 = K_{sp} \Rightarrow \text{Solubility} = \sqrt{K_{sp}}$$
$$= 10^{-5} M.$$

(*) Electrolytic Cell :

A non-spontaneous reaction is made to proceed by providing electrical energy from external source (battery)

$$A \longrightarrow B + C \quad \text{(Electrolytic decomposition)}$$

$\Delta G = +ve$

$E = -ve$

(eg. -1.0 volt)

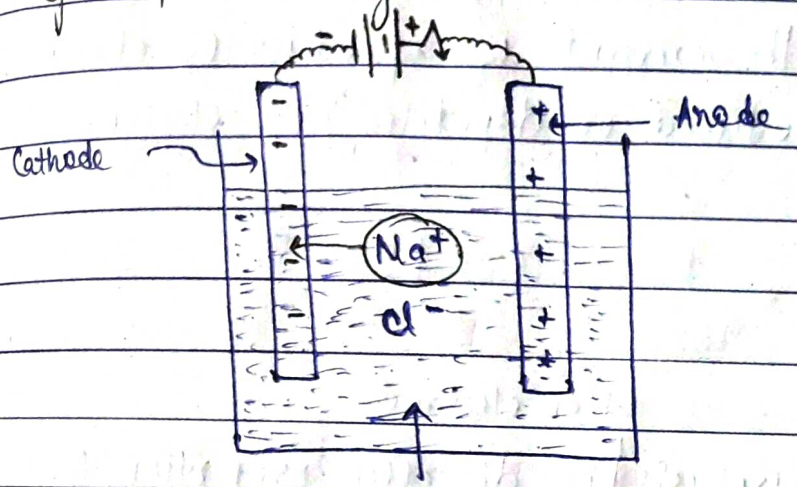
→ To move this non-spontaneous reaction forward, we have to apply a Potential difference of $\geq$ 1.0 volt from battery.

→ Electrolysis :

Chemical decomposition of an electrolyte with the help of electricity supplied from battery.

(Electrolyte maybe in aq. solution form or in Molten State.)

eg. An electrolytic cell



Electrolytic Cell

At cathode : (Reduction)

$Na^+ + e^- \longrightarrow Na$

At Anode : (oxidation)

$Cl^- \longrightarrow \frac{1}{2} Cl_2 \uparrow + e^-$

Cell Rkn :- $Na^+ + Cl^- \longrightarrow Na + \frac{1}{2} Cl_2 \uparrow$

Cell Rkn.
(ΔG = +ve)

* Na metal and Cl_2 gas are called products of electrolysis.

Sign : Cathode : -ve
Anode : +ve

} This is exactly reverse of sign of electrodes in Galvanic Cell.

always remember, in both Galvanic cell and Electrolytic cell, cathode is where reduction takes place and anode is where oxidation takes place (both have opposite signs in both these cases)

→ Some electrodes (Metal Rodes / Plates etc) may be inert (eg. Pt, graphite) or active (Cu, / Ni etc)

(*) Faraday's laws of Electrolysis :

→ These laws are about the amount of product of electrolysis (or mass of substance produced or depesited) on electrodes.

* Faraday's 1st law :-

Mass of product of electrolysis at an electrode is directly proportion to the quantity of electricity (i.e charge) supplied from battery

$W \propto q$ $\quad$ q = charge

$W \propto i \times t$ $\quad$ i = current

$W = z \times i \times t$ $\quad$ t = time period.

z = Electrochemical Equivalent (ECE) of substance.

W = mass of substance produced/ depesited.

$\Rightarrow z = W/q$

if $q = 1$ coulomb $\Rightarrow z = W$

⇒ ECE of a substance is defined as the mass of product of electrolysis when 1 coulomb charge is supplied to electrolytic cell.

⇒ Charge on 1 mol of e^-

$= 6.022 \times 10^{23} \times 1.6 \times 10^{-19}$ Coulomb

1 faraday (or 1F) $\simeq$ 96,500 Coulomb.

→ 1'F' is the charge present on 1 mol of e^-

1'F' = 96,500 coulomb.

⇒ 1'F' of electricity produces mass of a substance equal to its equivalent mass (E)

OR 1'F' electricity produces/ deposits amount of substance equal to 1g equivalent.

⇒ No. of faraday of electricity
= No. of gram eq. of substance
= No. of moles × Valency factor

} V-V imp.

⇒ Equivalent weight = $\frac{\text{Mol. wt}}{Vf}$

or $E = \frac{M}{Vf}$

Eg. $\underset{1\,mol}{Al^{3+}} + \underset{3'F'}{3e^-} \longrightarrow Al(s)$, Vf of $Al^{3+} = 3$

1'F' of electricity produces = $\frac{1}{3}$ mol of Al(s)

→ 1'F' (or 96,500 'c') electricity produces = Eg

∴ 1 'c' ——— = $\frac{E}{96500}$ ECE (z)

⇒ $W = z \times i \times t$

⇒ $\boxed{W = \frac{E}{96500} \times i \times t \times \eta}$ Remember!

where η = current efficiency
↑ 'eta'
(If η is 0.8 then it means current eff. is 80%)

Formulae's for Electrolysis :

$$\boxed{W = \frac{E}{96500} \times i \times t \times \eta}$$

$$\boxed{\text{No. of 'F'} = \text{No. of gram eq.} = \text{No. of moles} \times \text{v.f}}$$

gmp

i.e. charge

⇒ Equal quantity of electricity passes at cathode and anode if current effeciency of cathode & anode are equal.

∴ No. of gram equivalent of product at cathode

= No. of gram equivalent of product at anode.

* Faraday's 2nd law : it is a direct consequence of the 1st law, but still given as a law nonetheless

If equal charge passes into two electrolytes (eg if connected in series) then the ratio of mass of product of electrolysis in different electrolytes will be equal to the ratio of their eq. weight

$$W = \frac{E}{96500} \times q$$

If q is same, then $\frac{W_1}{W_2} = \frac{E_1}{E_2}$ → Faraday's Second law of Electrolysis.

⇒ Coulometer : A device (an electrolytic cell) which is used to determine amount of product of electrolysis.

Q. Calculate no. of faraday (electricity) needed for each of the following processes.

(i) 20g of SO_3^{2-} to SO_4^{2-}

(ii) 13.5g of $Al^{3+} \longrightarrow Al(s)$

(iii) Wg of nitrobenzene $\longrightarrow$ Aniline

(iv) 4g of methane $\longrightarrow$ $CO_2(g)$

Soln (i) $\overset{(+4)}{SO_3^{2-}} \longrightarrow \overset{(+6)}{SO_3^{2-}} + 2e^-$

$Vf = 2$

No. of 'F' = No. of gram eq. = No. of moles × V.f

$= \left(\frac{20}{80}\right) \times 2 = 0.5\ 'F'$

(ii) $Al^{3+} + 3e^- \longrightarrow Al$

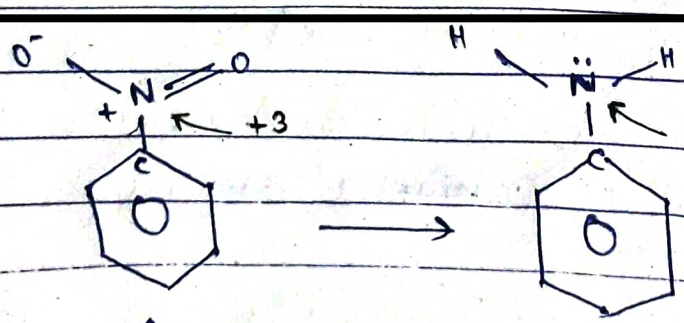
$Vf = 3$ $\qquad$ $Vf = 3$

$n_{Al}\left(\frac{13.5}{27}\right) = 0.5$ mol

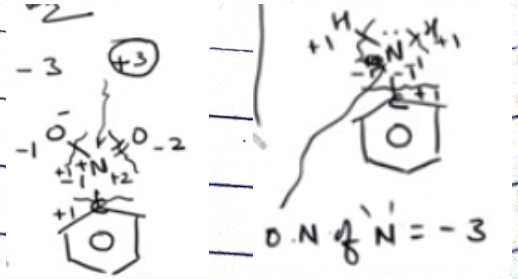
No. of 'F' = 0.5×3

$= 1.5\ 'F'$

(iii)



$Vf = 6$



No. of 'F' = $\left(\frac{W}{77+14+32}\right) \times 6$

$= \frac{W}{123} \times 6 = \frac{2}{41} W$

(iv) $\overset{(-4)}{CH_4} \longrightarrow \overset{(+4)}{CO_2} + 8e^-$

$\uparrow$

$Vf = 8$

No. of 'F' = $\frac{4}{16} \times 8 = 2F$

Q. What mass of copper will be deposited on an inert electrode if a current of 96.5 ampere is supplied for 1 hr to an aq. $CuSO_4$ soln (excess)

Soln $i = 96.5$ amp, $t = 60 \times 60$ sec

$Cu^{2+} + 2e^- \longrightarrow Cu$

$W = \frac{E}{96500} \times i \times t \times \eta$

$= \frac{64}{2 \times 96500} \times 96.5 \times 60 \times 60$

$= 32 \times 6 \times 0.6$

$= 32 \times 3.6$ g Ans.

Q. During electrolysis of water, 50 ml of H_2 gas is evolved at an inert electrode. In same time period, what volume of O_2 gas will be evolved under same P & T conditions.

Rkn. at electrodes:

Cathode: $2H_2O(l) + 2e^- \longrightarrow H_2(g) + 2OH^-(aq)$

2F 1 mol

Anode: $2H_2O(l) \longrightarrow 4H^+(aq) + O_2\uparrow + 4e^-$

1 mol 4F

0.25 mol 1F

Soln:

0.5 mol $H_2\uparrow$, 0.25 mol $O_2\uparrow$ when 1F electricity is passed

V_{O_2} liberated = 25 ml Ans for given amount of electricity

* Equal quantity of electricity (charge) passes at cathode and anode.

Q Imp For what time, a current of 2 amp to be passed to coat Ag metal on a metal object of surface area 100 cm^2 with Ag coating thickness of 10^{-3} mm.

$d_{Ag(s)} = 10.8$ g/cc, Ag → 108

Soln $Ag^+ + e^- \longrightarrow Ag(s)$

V_{Ag} coated on Metal = 100 $cm^2 \times 10^{-4}$ cm

= 10^{-2} cm^3

$W_{Ag} = 10.8 \times 10^{-2}\ g$

$n_{Ag} = \frac{10.8 \times 10^{-2}}{108} = 10^{-3}\ mol$

$Ag^+ + e^- \longrightarrow Ag$

$10^{-3}\ F$ $\qquad$ $10^{-3}\ mol$

No. of 'F' $= \frac{i \times t}{96500} = \frac{2 \times t}{96500} = 10^{-3}$

$t = \frac{96.5}{2} = 48.25\ sec$ Ans.

Alternatively

$W = \frac{E}{96500} \times i \times t \times n$

$10.8 \times 10^{-2} = \frac{108}{1 \times 96500} \times 2 \times t \times 1$

$t = 48.25\ sec.$

Q. Three coulometers consisting of $AgNO_3$, ~~$ZnCl_2$(aq)~~ $CuCl_2$(aq), $AuCl_3$(aq) are electrolysed by supplying 1 'F' of electricity. Find ratio of masses of Ag, Cu, Au obtained

(i) if connected in Series.
(ii) if connected in parallel.
and charge passen in 1:2:3 ratio.

E Ag → 108
Cu → 64
Au → 197

Soln (i) q- same,

$$W_{Ag} : W_{Cu} : W_{Au} = E_1 : E_2 : E_3$$

$$= \frac{108}{1} : \frac{64}{2} : \frac{197}{3}$$

(ii) $W_1 : W_2 : W_3 = E_1q_1 : E_2q_2 : E_3q_3$

$$= \frac{108}{1} \times \frac{1}{6} : \frac{64}{2} \times \frac{2}{6} : \frac{197}{3} \times \frac{3}{6}$$ Ans.

(*) Preferential Electrolysis :
(or Preferential Deposition)

→ Particularly when electrolysis of aq. solution of electrolytes is carried out, there may be two or more species present in solution which may compete for reduction at cathode or oxidation at anode.

→ For species competing for Reduction at Cathode, higher is the discharge potential (or reduction potential or SRP), greater will be its ease to undergo reduction at cathode.

→ In general, cations undergo reduction at Cathode,

→ Based on discharge potential (or RP or SRP),

K^+, Ca^{2+}, Na^+, Mg^{2+}, Al^{3+}, $H_2O(l)$, Zn^{2+}, Fe^{2+}, H^+, Cu^{2+}, Ag^+, Au^{3+} →

increasing order of discharge potential
(or increasing order of ease of reduction at Cathode)

$2H_2O(l) + 2e^- \longrightarrow \underset{\uparrow v.f = 2}{H_2(g)} + 2OH^-_{(1M)}$, $E^\circ_{SRP} = -0.83$ Volt

↳ At $pH = 7$

$E_{RP} = -0.42$ Volt.

$2H^+ + e^- \longrightarrow \underset{\uparrow v.f = 2}{H_2\uparrow_{(g)}}$, $E^\circ_{SRP} = 0$

→ Based on discharge potential, for species competing at anode,

SO_4^{2-}, NO_3^-, $H_2O(l)$, OH^-, Cl^-, Br^-, I^- ⟶

increasing tendency or ease to undergo oxidation at anode.

(or in general increasing order of discharge potential)

$2H_2O(l) \longrightarrow O_2(g) + 4H^+ + 4e^-$, $E^\circ_{SOP} = -1.23$ Volt

↳ E_{OP} at $pH = 7 = -0.82$ Volt

$2OH^- \longrightarrow \frac{1}{2}O_2 \uparrow + H_2O(l) + 2e^-$

↳ $E^\circ_{SOP} = -0.4$ Volt.

* At anode, In general

OP ↑ (or SOP ↑), ease of oxidation ↑

NOTE :

(1) aq. NaCl Solution $\xrightarrow{\text{Electrolysis}}$

Cathode : $2H_2O(l) + 2e^- \longrightarrow H_2\uparrow + 2OH^-$ thus, nature of soln. becomes basic

Anode : $2Cl^- \longrightarrow Cl_2\uparrow + 2e^-$

See this : $2Cl^- \longrightarrow Cl_2\uparrow + 2e^-$, $E^\circ_{SOP} = -1.35$ Volt

$2H_2O(l) \longrightarrow O_2\uparrow + 4H^+ + 4e^-$, $E^\circ_{SOP} = -1.23$ Volt

According to this data, Electrolysis of aq. NaCl (Brine) should liberate O_2 gas (from oxidation of water) in preference to Cl_2 gas (from oxidation of Cl^-)

⇒ But the fact is that Cl_2 liberates in preference to O_2 because liberation of O_2 gas from oxidation of water is kinetically slow process, though it is thermodynamically favoured.

$2H_2O + 2e^- \xrightarrow[\text{(Reduction)}]{\text{Cathode}} H_2\uparrow + 2OH^-$, $E^\circ_{Red} = -0.83\,V$

$2Cl^- \longrightarrow Cl_2\uparrow + 2e^-$, $E^\circ_{SOP} = -1.35\,V$

$2H_2O + 2Cl^- \longrightarrow H_2\uparrow + 2OH^- + Cl_2\uparrow$, $E^\circ_{cell} = E^\circ_{Oxd} + E^\circ_{Red}$

$= -0.83 - 1.35$

$= -2.18$ Volt

non-spontaneous

$2H_2O + 2e^- \longrightarrow H_2\uparrow + 2OH^-$, $E^\circ_{SRP} = -0.83\ V$

$4H_2O \longrightarrow \frac{1}{2}O_2\uparrow + 2H^+ + 2e^-$, $E^\circ_{SOP} = -1.23\ V$

$H_2O \rightarrow H_2\uparrow + \frac{1}{2}O_2\uparrow$, $E^\circ_{cell} = -0.83 - 1.23$

$= -2.06$ Volt

thermodynamic data suggests liberation of oxygen is more favourable, but in reality, the observations suggest otherwise.

⇒ for effective liberation of O_2 gas from oxidation of H_2O, an additional voltage is required to be applied and this extra voltage required is called 'Oversvoltage' or 'Overpotential'

Imp → In other words, oversvoltage is considerably high for O_2 gas (from oxidation of H_2O) and oversvoltage is negligible for $Cl_2\uparrow$ (from oxidation of Cl^-)

thus, two major examples of over-potential : in electrolysis of brine and in production of NaOH in Castner-Kellner cell

V.Imp → In production of NaOH by Castener - Kellener cell (→ S- Block), Na^+ are reduced on Hg(l) cathode in preference to liberation of $H_2(g)$ from H_2O.

AIIMS → This means that oversvoltage of liberation of H_2 gas from reduction of water is considerably high on Hg(l) cathode.

(2) aq. H_2SO_4 $\xrightarrow{\text{Electrolysis}}$ Cathode : $2H^+ + 2e^- \longrightarrow H_2\uparrow$

Anode : $SO_4^{2-} \longrightarrow X$

$2H_2O(l) \longrightarrow O_2\uparrow + 4H^+ + 4e^-$

conc. H_2SO_4 or $(NH_4)_2SO_4$ → Electrolysis

Cathode : $2H^+ + 2e^- \longrightarrow H_2\uparrow$

or

$2NH_4^+ + 2e^- \longrightarrow H_2\uparrow + 2NH_3\uparrow$

Anode : $2SO_4^{2-} \longrightarrow S_2O_8^{2-} + 2e^-$

↑ anion of Marshall's Acid.
(Peroxydisulphate ion)

$H_2S_2O_8$ (Marshall's acid)

↑ Peroxydisulphuric acid

* Products of electrolysis of aq. Solution :

aq. Soln	Electrode	Rkn at Cathode	Rkn. at Anode
(1) aq. NaCl	Pt (inert)	$2H_2O + 2e^- \rightarrow H_2\uparrow + 2OH^-$	$2Cl^- \rightarrow Cl_2\uparrow + 2e^-$
(2) aq. $CuSO_4$	Pt	$Cu^{2+} + 2e^- \rightarrow Cu(s)$	$2H_2O(l) \rightarrow O_2\uparrow + 4H^+ + 4e^-$
(3) aq. Na_2SO_4	Pt	$2H_2O + 2e^- \rightarrow H_2\uparrow + 2OH^-$	$2H_2O \rightarrow O_2\uparrow + 4H^+ + 4e^-$
(4) CH_3COONa (Kolbe's Electrolysis)	Pt	$2H_2O + 2e^- \rightarrow H_2\uparrow + 2OH^-$	$2CH_3COO^- \rightarrow C_2H_6(g) + 2CO_2(g) + 2e^-$

$HCOOK_{(aq)}$	Pt	$2H_2O + 2e^- \longrightarrow H_2\uparrow + 2OH^-$	$2HCOO^- \longrightarrow H_2\uparrow + 2CO_2\uparrow + 2e^-$
(5) $CuSO_{4(aq)}$	Cu electrodes		

(5) ⇒ It is a Special Case called 'attackable electrode' (Cu rods/plates may behave as active electrodes)

* Possibilites at Cathode : (Cu Metal)

$Cu^{2+} + 2e^- \longrightarrow Cu(s)$, $E^\circ_{SRP} = 0.34V$ ← This reaction occurs at cathode.

$2H_2O(l) + 2e^- \longrightarrow H_2\uparrow + 2OH^-$, $E^\circ_{SRP} = -0.83V$

* Possibilites at Anode : (Cu Metal)

$2H_2O(l) \longrightarrow O_2\uparrow + 4H^+ + 4e^-$, $E^\circ_{SOP} = -1.23V$

✗ $2SO_4^{2-} \longrightarrow S_2O_8^{2-} + 2e^-$, E°_{SOP} = highly -ve

$Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$, $E^\circ_{SOP} = -0.34V$

↓ This reaction occurs at anode.

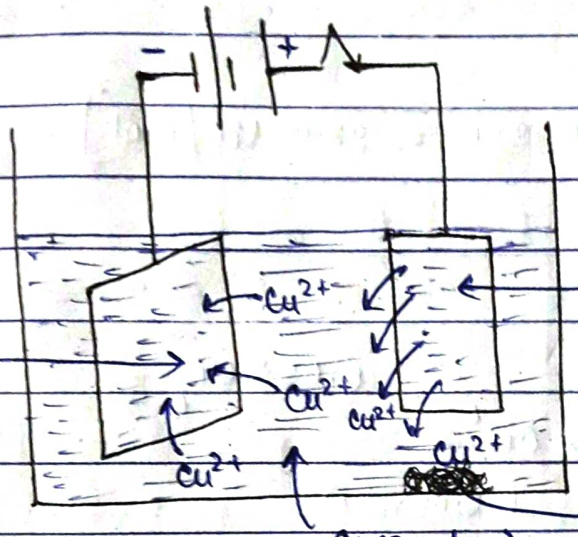
⇒ Net Reaction, Cathode : $Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu(aq)$

Anode : $Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$

* Application of this concept is used in Refining of metals like Cu, Zn, Sn etc called ELECTROREFINING.

⇒ Electrorefining of Copper :-

→ Blister Copper (≈ 98 to 99 % pure Cu) is purified by this method. the obtained pure Cu is mainly used in electrical wires



[Active Electrode]

Impure Cu Rod (Anode)

Pure Cu plate (cathode)

By-product

Anode mud
(Non-oxidizable impurities)
eg. Ag, Au, Se, As etc.

eg. $Ag \longrightarrow Ag^+ + e^-$, $E^\circ_{SOP} = -0.79V$

$2H_2O(l) \longrightarrow O_2 \uparrow + 4H^+ + 4e^-$, $E^\circ_{SOP} = -1.23V$

Q. 1 lit. of 0.1M aq. $CuSO_4$ is electrolysed using Pt electrodes. Assuming volume of solution to remain constant.

(a) Find pH, Volume of gas(es) evolved at STP if 0.1F of electricity is supplied

(b) Find pH, volume of gas(es) evolved at STP if 0.4F of electricity is supplied.

[Take Molar Volume of gas at STP = 22.4 lit/mol]

Soln : 1 lit., $CuSO_4$ (0.1M)

(a) Cathode : $Cu^{2+} + 2e^- \longrightarrow Cu$ ← 0.05 × 64 g

0.1 mol − 0.05 | 0.1F (LR) | (0.05 mol)

Anode : $2H_2O(l) \longrightarrow O_2\uparrow + 4H^+ + 4e^-$

$\frac{1}{4} \times 0.1$ mol | 0.1 mol | 0.1F

$[H^+]_f = \frac{0.1}{1} = 0.1M$

pH = 1 Ans, $(V_{O_2})_{STP} = \frac{1}{4} \times 0.1 \times 22.4$

$= 0.56$ lit Ans.

(b) 0.4 'F' electricity supplied

At cathode : $Cu^{2+} + 2e^- \longrightarrow Cu(s)$

0.1 mol (LR) | 0.4F − 0.2F = 0.2F | ↑ 0.1 mol

$2H_2O + 2e^- \longrightarrow H_2\uparrow + 2OH^-$

(0.2F) ↑ 0.1 mol ↑ 0.2 mol

At anode : $2H_2O(l) \longrightarrow O_2\uparrow + 4H^+ + 4e^-$

0.1 mol 0.4 mol (0.4F)

V gases at STP $= (0.1 + 0.1) \times 22.4$

$= 4.48$ lit.

$H^+ + OH^- \longrightarrow H_2O(l)$

0.4 mol 0.2 mol

- 0.2 mol

= 0.2 mol

$[H^+]_f = \frac{0.2}{1} = 0.2\,M$

$pH = 1 - 0.3$

$= 0.7$ Ans.

Q During preparation of Marshall's acid ($H_2S_2O_8$), O_2 gas is also released at anode as by-product. What are the no. of moles of $H_2S_2O_8$ produced if 8.96 lit. of H_2 at cathode and 2.24 lit of O_2 at anode at STP are formed.

Sol^n : Cathode : $2H_2O + 2e^- \longrightarrow H_2\uparrow + 2OH^-$

↑ $v_f = 2$

Anode : $2H_2SO_4 \longrightarrow H_2S_2O_8 + 2H^+ + 2e^-$

↑ $v_f = 2$

$2H_2O \longrightarrow O_2\uparrow + 4H^+ + 4e^-$

↑ $v_f = 4$

electricity consumed in cathode = electricity consumed in anode

as no. of gm. equivalents = no. of Faraday of e-

No. of gram equivalent of products at Cathode

= No. of gram eq. of products at anode

gram eq. of H_2 = gram eq. of $H_2S_2O_8$ + gram eq. of O_2

$$\frac{8.96}{22.4} \times 2 = n \times 2 + \frac{2.24}{22.4} \times 4$$

$$2n + 0.4 = 0.8$$

$n = 0.2$ mol of Marshall's acid

3. 1 lit of 0.5M Sodium acetate is electrolysed using Pt electrodes until entire sodium acetate is exhausted. Find total volume of gases evolved at STP. [Molar vol. of gas STP = 22.4 lit/mol]

Soln Kolbe's electrolysis

Cathode : $2H_2O + 2e^- \longrightarrow H_2\uparrow + 2OH^-$

(0.5F) (0.25 mol)

Anode : $2CH_3COO^- \longrightarrow C_2H_6\uparrow + 2CO_2\uparrow + 2e^-$

(0.5 mol) (0.25 mol) (0.5 mol) (0.5F)

$(n_T)_{gases}$ = 0.25 + 0.25 + 0.5

= 1 mol

$(V_{gases})_{STP}$ = 1 × 22.4 lit Ans

Q. Cd- amalgam (Cd-Hg) is prepared by electrolysis of $CdCl_2$ (aq) using Hg(l) cathode. How long a current of 9.65 amp to be passed in order to prepare 10% Cd-Hg amalgam on cathode of Hg (2g)

(Cd → 112)

Soln: $Cd^{2+} + 2e^- \longrightarrow \underset{x\,g}{Cd} \xrightarrow{Hg(l)} \underset{(2+x)\,g}{Cd-Hg}$

$x\,g$ = 10% of $(2+x)\,g$

$x = 0.1(2+x) = 0.2 + 0.1x$

$0.9x = 0.2 \Rightarrow x = 2/9\ g$

$$\frac{2}{9} = \frac{112}{2 \times 96500} \times 9.65 \times t \times 1$$

$$\Rightarrow \boxed{t = \frac{4 \times 10^4}{9 \times 112}\ \text{sec.}}$$ Ans.

(*) - Electrolytic Conduction -

Metallic or electronic Conduction.	Electrolytic Conduction
(1) This is due to the movement of free e^- (or valence e^-) in metals under electric field.	(1) It is due to movement of ions under electric field in aq. solution or molten state of an electrolyte.
→ Composition of metal does not change due to conduction.	→ Composition of an electrolyte soln. may change due to electrolysis.
(2) Metallic conduction depends on nature of metal (eg. no. of valence e^- per metal atom, crystal structure etc), temperature, length & area of cross section.	(2) It depends on conc., nature of electrolyte (Strong / weak), nature of solvent & its viscosity temperature.
(3) T↑, metallic conduction ↓ (on increase in temp., vibration of metal kernels become more vigorous which hinders the movement of free e^-, under electric field.	(3) T↑, Viscosity ↓, electrolytic conduction ↑

$$R \propto \frac{l}{A}$$

$$R = \rho \times \frac{l}{A}$$

R → Resistance (ohm, Ω)

ρ → Resistivity (or Specific Resistance) (Ω-cm)

$$\frac{1}{R} = \frac{1}{\rho} \times \frac{A}{l}$$

$$\frac{1}{\rho} = \frac{1}{R} \times \frac{l}{A}$$

$\frac{1}{\rho}$ → Conductivity (or Specific conductance) (κ) → Kappa

$\frac{1}{R}$ → conductance 'G'

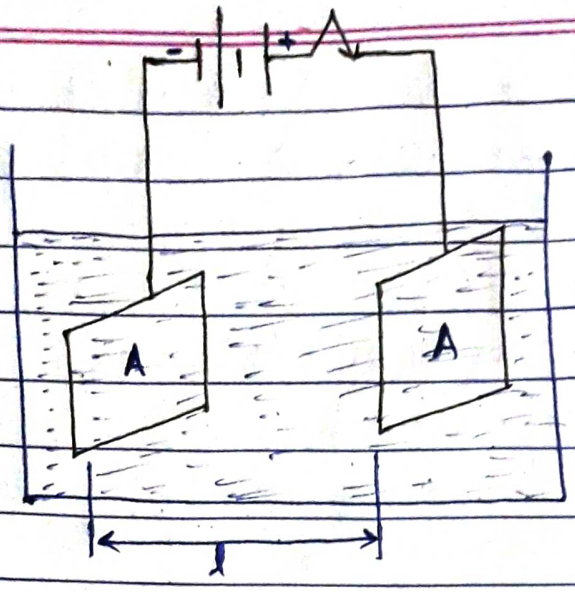
$$\kappa = G \times \frac{l}{A}, \quad \frac{l}{A} = \text{Cell constant (unit : } cm^{-1} \text{ or } m^{-1})$$

κ → $\Omega^{-1} cm^{-1}$ or $S cm^{-1}$ (CGS) or $S m^{-1}$ (SI)

G → Ω^{-1} or 'S'

l = separation between parallel electrodes.

A = Area of cross section of electrodes (or Plates).



$$K = G \times \frac{l}{A}$$

A = Common projected area between plates dipping in solution.

NOTE

(1) $$K = G \times \frac{l}{A} \times \frac{l}{l}$$

$A \times L = 'V'$ (Volume of solution between parallel plates)

$l = 1cm$, $V = 1cm^3$

$$K = G$$

V. Imp

→ Conductivity(K) is the conductance of unit volume of solution, filled between two parallel plates at unit distance.

(2) Resistance of a solution can be experimentally measured using wheat-stone bridge in a conductivity cell.

From measured resistance, conductivity (K) can be determined.

or 1/resistance

→ like in metallic (Cu) wire, G → changes with length & area of cross-section.

or 1/resistivity

but k is constant at a given temperature.

→ In aq. soln of electrolyte, G → changes with l and A.

but k is constant. k depends on conc. of solution and temp.

V. Imp

(3) Conductivity is an additive property.

$$k_{total} = k_1 + k_2 + k_3 + \dots + k_{water}$$

$$G_{total} \times \frac{l}{A} = G_1 \times \frac{l}{A} + G_2 \times \frac{l}{A} + \dots + G_{water} \times \frac{l}{A}$$

$$G_{total} = G_1 + G_2 + \dots + G_{water}$$

$$\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_{water}}$$

All electrolytes in solution are connected in parallel.

→ k_{water} (conductivity of pure water (distill water) is very low and may be ignored in questions if data is not given.

Λ – Bada lambda

λ – Chota lambda.

λ_{Na^+} , λ_{Cl^-}

Λ_m (NaCl)

(*) – Molar Conductivity – (Λ_m)
(Molar Conductance)

→ It is the conductance of the solution which contains 1 mol of electrolyte filled between parallel electrodes of unit distance apart and large enough to accomodate complete volume.

$$K = G \times \frac{l}{A}$$

$$G = K \times \frac{A}{l} \times \frac{l}{l}$$

$$G = K \times \frac{V}{l^2}$$

for the definition of Molar conductivity,
$l = 1$ and soln contains 1 mol of electrolyte.

$$\Lambda_m = G$$

IN CGS:

$$\Lambda_m = K \times V$$

V : cm^3/mol — Volume of solution which contains 1 mol of electrolyte.

Λ_m : $Scm^2 mol^{-1}$

K : Scm^{-1}

In SI:

$$\Lambda_m = K \times V$$

V : m^3/mol (SI)

Λ_m : $Sm^2 mol^{-1}$

K : Sm^{-1}

Students, Plz do not use them as formula.

In CGS unit : Conc. of solution = c mol/lit

∵ c mol electrolyte —— = 10^3 cm^3

∴ 1 mol electrolyte —— = $\frac{1000}{c}$ $cm^3/mol = V$

$\Lambda_m = K \times V$

$= K \times \frac{1000}{c}$

$\Lambda_m = \frac{1000 \times K}{Molarity}$ → in S/cm

→ In CGS

Remember!

in S/cm^2 mol

In SI unit : Conc. = c mol/lit

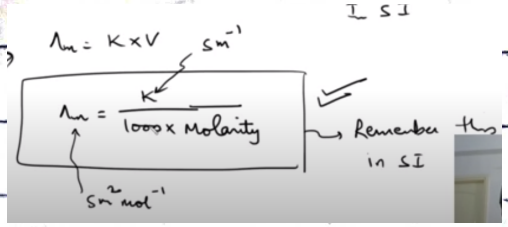
∵ c mol of electrolyte —— = 10^{-3} m^3

∴ 1 mol of electrolyte —— = $\frac{1}{1000 \times c}$ $m^3/mol = V$

$\Lambda_m = K \times V$

$\Lambda_m = \frac{K}{1000 \times Molarity}$

Remember!



* Equivalent Conductivity (Λ_{eq}) :
(Equivalent Conductance)

Conductance of solution which contains 1g eq. of electrolyte

$$\Lambda_{eq} = \frac{1000 \times K}{\text{Normality}}$$

K : Scm^{-1} → In CGS

Λ_{eq} : $Scm^2 eq^{-1}$

$$\Lambda_{eq} = \frac{K}{1000 \times \text{Normality}}$$

K : Sm^{-1} → In SI unit

Λ_{eq} : $Sm^2 eq^{-1}$

* Relation between Λ_m and Λ_{eq} :

$$\frac{\Lambda_m}{\Lambda_{eq}} = \frac{1000K}{M} \times \frac{N}{1000K}$$

$$\Lambda_{eq} = \frac{\Lambda_m}{\text{Valency factor}}$$

V.f of NaCl = 1
of $BaSO_4$ = 2
of $Al_2(SO_4)_3$ = 6

Q. Calculate conductivity contribution of AgCl if conductivity of saturated aq. soln of AgCl is 10^{-6} S cm^{-1}
Given k (water) $= 5 \times 10^{-8}$ S cm^{-1}

Soln $\quad k_T = k_{AgCl} + k_{H_2O}$

$k_{AgCl} = 10^{-6} - 0.05 \times 10^{-6}$

$= 9.995 \times 10^{-6} \quad = 0.95 \times 10^{-6}$

$= 9.995 \times 10 \quad = 9.5 \times 10^{-7}$ S cm^{-1}

Q. Resistance of a conductivity cell filled with 0.1M KCl soln is 100 Ω. If the resistance of the same cell when filled with 0.02 M KCl soln. is 520 Ω. Calculate the conductivity cell constant and molar conductivity of 0.02 M KCl soln.

Given k (0.1M KCl soln) $= 1.3$ S m^{-1}

Soln $\quad k = \frac{1}{R} \times \frac{l}{A} \Rightarrow 1.3 = \frac{1}{100} \times \frac{l}{A}$

$\Rightarrow \frac{l}{A} = 130$ m^{-1} Ans.

(cell constant)

k (0.02M KCl) $= \frac{1}{520} \times 130 = \frac{1}{4} = 0.25$ S m^{-1}

Λ_m (0.02m KCl) $= \frac{k}{1000 \times c} = \frac{0.25}{1000 \times 0.02}$

$= \frac{0.25}{20} = 1.25 \times 10^{-2}$ S m^2 mol^{-1}.

(*) Variation of Conductivity (k) with Concentration or Dilution :

dilution ↑, (or conc. ↓), Conductivity ↓

→ Conductivity is the conductance of unit vol. of solution containing electrolyte.

If volume of solution is doubled, then no. of current carrying ions per unit vol. of soln decreases and conductance will decrease.

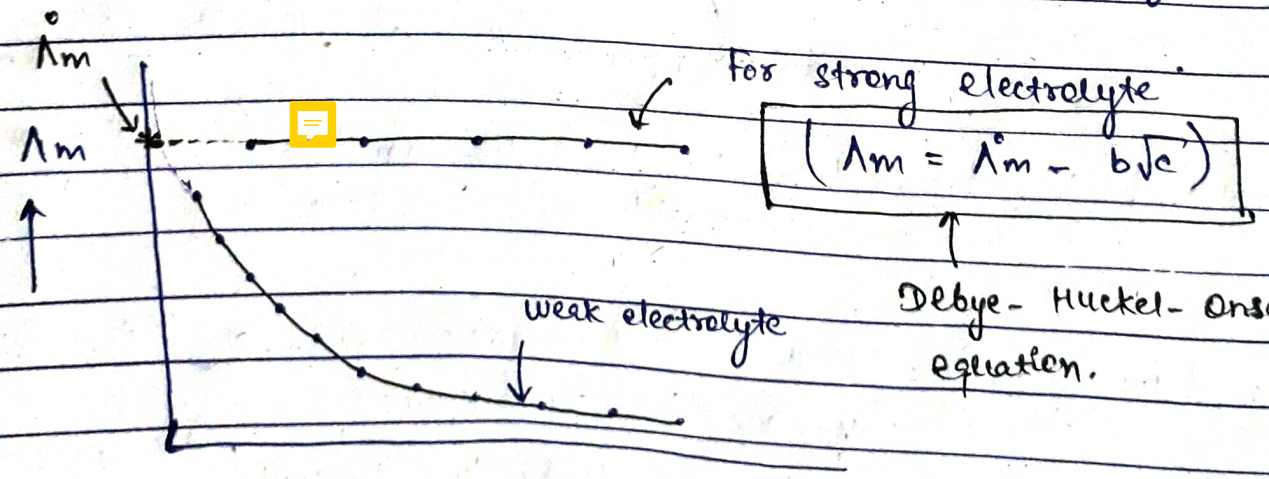
⇒ dilution ↑ (or conc. ↓), k ↓

(For strong as well as weak electrolytes)

* Variation of Λ_m (or Λ_{eq}) with conc.

→ dilution ↑ (or Conc. ↓), Λ_m or Λ_{eq} ↑

for both strong as well as weak electrolytes.



Experimental Variation of Λ_m with $\sqrt{c}$.

→ In case of weak electrolyte, Λ_m increases sharply as solution is diluted because no. of current carrying ions sharply increases on dilution (Ostwald's dilution law)

$$\alpha \approx \sqrt{\frac{K_a}{C}} \Rightarrow \alpha \propto \frac{1}{\sqrt{C}}$$

$\Lambda_m \propto \alpha$

$\Lambda_m \propto \frac{1}{\sqrt{C}}$, $\sqrt{C} \uparrow$, $\Lambda_m \downarrow$

$\sqrt{C} \downarrow$, $\Lambda_m \uparrow$

→ In case of strong electrolyte, Λ_m or Λ_{eq} gradually increases with dilution because the frictional resistance between hydrated ions are decreased on dilution.

$$\Lambda_m = \Lambda_m^\circ - b\sqrt{C} \quad \text{(for strong Electrolyte)}$$

Λ_m ⟶ Molar Conductivity at conc. 'c'

Λ_m° = limiting molar conductivity or molar conductivity at ∞ dilution.

C = Concentration of Solution.

b = Proportionality constant which depends on

→ Nature of electrolyte

* NaCl, KCl, $BaCl_2$
  1:1 1:1 1:2

* b is same for similar nature of electrolyte.
  i.e b is same for NaCl & KCl.

i.e 'b' is same for $BaCl_2$, $CaCl_2$

→ Viscosity of Solvent

Viscosity of Solvent ↑, b ↑

→ Temperature –

Temp. ↑, Viscosity ↓, b ↓

→ dielectric constant of Solvent ↑, b ↓

Note: In numerical problems, variation of Λ_m with concentration can be ignored if need arise, only for strong electrolyte.

eg. Λ_m(0.1M KCl) = Λ_m(0.2M KCl) [On 'Need']

→ Λ_m° for strong electrolyte can be experimentally determined by extrapolation. (as variation is linear)

But not for weak electrolyte.
(as variation is non-linear)

* This problem was solved by "Kohlraush."

* Kohlraush law of independent migration of ions :-

→ At ∞ dilution, molar conductivity of an electrolyte (strong or weak) is the sum of ionic conductivities (or molar conductivities) of constituent ions.

ionic conductivities

$$\Lambda^\circ_{m,NaCl} = \lambda^\circ_{m,Na^+} + \lambda^\circ_{m,Cl^-}$$

$$\Lambda^\circ_{m\ CH_3COOH} = \lambda^\circ_{m,CH_3COO^-} + \lambda^\circ_{m,H^+}$$

$$\boxed{\Lambda^\circ_{m,BaCl_2} = \lambda^\circ_{m,Ba^{2+}} + 2\times\lambda^\circ_{m,Cl^-}}$$

$$\Lambda^\circ_{eq} = \frac{\Lambda^\circ_m}{v\text{-f}}$$

$$\boxed{\Lambda^\circ_{eq,BaCl_2} = \frac{\Lambda^\circ_{m,BaCl_2}}{2} = \frac{1}{2}\lambda^\circ_{m,Ba^{2+}} + \lambda^\circ_{m,Cl^-}}$$

$$\Rightarrow \Lambda^\circ_{eq,BaCl_2} = \frac{1}{2}\times\left(\lambda^\circ_{eq,Ba^{2+}}\times 2\right) + \left(1\times\lambda^\circ_{eq,Cl^-}\times 1\right)$$

$$\Rightarrow \boxed{\Lambda^\circ_{eq,BaCl_2} = \lambda^\circ_{eq,Ba^{2+}} + \lambda^\circ_{eq,Cl^-}}$$

eg. $Al_2(SO_4)_3$

$$\Lambda^\circ_{m,Al_2(SO_4)_3} = 2\times\lambda^\circ_{m,Al^{3+}} + 3\times\lambda^\circ_{m,SO_4^{2-}}$$

$$\Lambda^\circ_{eq,Al_2(SO_4)_3} = \frac{1}{3}\lambda^\circ_{m,Al^{3+}} + \frac{1}{2}\lambda^\circ_{m,SO_4^{2-}}$$

$$= \frac{1}{3}\times\left(3\times\lambda^\circ_{eq,Al^{3+}}\right) + \frac{1}{2}\times\left(2\times\lambda^\circ_{eq,SO_4^{2-}}\right)$$

$$\Rightarrow \boxed{\Lambda^\circ_{eq,\ Al_2(SO_4)_3} = \lambda^\circ_{eq,\ Al^{3+}} + \lambda^\circ_{eq\ ,\ SO_4^{2-}}}$$

General case :

$$A_xB_{y(aq)} \longrightarrow xA^{y+}_{(aq)} + yB^{x-}_{(aq)}$$

$$\Lambda^\circ_{m\ ,\ A_xB_{y(aq)}} = x \times \lambda^\circ_{m,\ A^{y+}} + y \times \lambda^\circ_{m\ ,\ B^{x-}}$$

$$\Lambda^\circ_{eq\ ,\ A_xB_y} = \frac{1}{y}\lambda^\circ_{m\ ,\ A^{y+}} + \frac{1}{x}\lambda^\circ_{m\ ,\ B^{x-}}$$

$$\Lambda^\circ_{eq\ ,\ A_xB_y} = \frac{1}{y} \times y \times \lambda^\circ_{eq\ ,\ A^{y+}} + \frac{1}{x} \times x \times \lambda^\circ_{eq\ ,\ B^{x-}}$$

$$\boxed{\Lambda^\circ_{eq,\ A_xB_y} = \lambda^\circ_{eq\ ,\ A^{y+}} + \lambda^\circ_{eq\ ,\ B^{x-}}}$$

→ Kohlraush law is based on experimental observation.

eg. Cation is Same,

$$\Lambda^\circ_{m\ ,KCl} - \Lambda^\circ_{m\ ,KF} = \Lambda^\circ_{m\ ,LiCl} - \Lambda^\circ_{m\ ,LiF}$$

eg. anion is same

$$\Lambda^\circ_{m\ ,LiBr} - \Lambda^\circ_{m\ KBr} = \Lambda^\circ_{m\ LiF} - \Lambda^\circ_{m\ KF}$$

Kohlrausch law :

At ∞ dilution, there is a definite contribution of constituent ions in molar conductivity of electrolyte.

$$\Lambda_m^\circ, KCl = \lambda_{m}^\circ, K^+ + \lambda_{m}^\circ, Cl^-$$

* Applications of Kohlrausch's law :-

1) To determine Λ_m° or Λ_{eq}° of weak electrolyte.

eg. to determine, $\Lambda_m^\circ, CH_3COOH$

Given : $\underline{\Lambda_m^\circ HCl, \Lambda_m^\circ NaCl, \Lambda_m^\circ, CH_3COONa}$

all strong electrolyte

$$\Lambda_m^\circ, CH_3COOH = \lambda_{m}^\circ, CH_3COO^- + \lambda_{m}^\circ, H^+$$

$$\Lambda_m^\circ, HCl = \lambda_{m}^\circ, H^+ + \lambda_{m}^\circ, Cl^- \quad -(1)$$

$$\Lambda_m^\circ, NaCl = \lambda_{m}^\circ, Na^+ + \lambda_{m}^\circ, Cl^- \quad -(2)$$

$$\Lambda_m^\circ, CH_3COONa = \lambda_{m}^\circ, CH_3COO^- + \lambda_{m}^\circ Na^+ \quad -(3)$$

(1) + (3) − (2)

$$\Lambda_m^\circ, HCl + \Lambda_m^\circ, CH_3COONa - \Lambda_m^\circ, NaCl = \lambda_{m}^\circ, H^+ + \lambda_{m}^\circ, CH_3COO^-$$

$$\boxed{\Lambda_m^\circ, CH_3COOH = \Lambda_m^\circ, HCl + \Lambda_m^\circ CH_3COONa - \Lambda_m^\circ, NaCl}$$ Ans

Q. Λ_m°, $BaCl_2 = x$

Λ_m°, $Ba(OH)_2 = y$, Λ_m°, $NH_4Cl = z$

Find Λ_m°, NH_4OH

Soln : $\Lambda_m^\circ, NH_4OH = \left(z + \frac{y}{2} - \frac{x}{2}\right)$ Ans.

(2) To determine degree of ~~dis~~ ionisation (α) and $K_{diss.}$ of a weak electrolyte :-

eg. Weak acid

$$HA(aq) \rightleftharpoons H^+(aq) + A^-(aq)$$

At start,	c	0	0
At eqm,	$c(1-\alpha)$	$c\alpha$	$c\alpha$

$\Lambda_m, c \propto$ degree of ~~diss~~ ionisation

$\Lambda_{m,c} = \Lambda_m^\circ \times \alpha \Rightarrow$ $\boxed{\alpha = \frac{\Lambda_{m,c}}{\Lambda_m^\circ}}$

↑ limiting molar conductivity.

$$K_a(HA) = \frac{c\alpha^2}{1-\alpha} = \frac{c \times \frac{\Lambda_m^2}{\Lambda_m^{\circ 2}}}{\left(1 - \frac{\Lambda_m}{\Lambda_m^\circ}\right)}$$

Q. Find K_b of 0.1 M NH_4OH if Λ_m, 0.1M NH_4OH = 5 $scm^2 mol^{-1}$ and Λ_m°, NH_4OH = 500 $scm^2 mol^{-1}$

Soln:

$$NH_4OH\,(aq) \rightleftharpoons NH_4^+\,(aq) + OH^-\,(aq)$$

$$c(1-\alpha) \qquad c\alpha \qquad c\alpha$$

$$\alpha = \frac{\Lambda_m}{\Lambda_m^\circ} = \frac{5}{500} = 10^{-2}$$

$$K_a = \frac{c\alpha^2}{1-\alpha} = c\alpha^2 = 0.1 \times 10^{-4} = 10^{-5}$$

(3) To determine K_{sp} of a sparingly soluble salt:

Q. The conductivity of saturated solution of $BaSO_4$ is 3.06×10^{-6} scm^{-1}. Find K_{sp} of $BaSO_4$ if Λ°_{eq}, $BaSO_4$ = 150 $scm^2 eq^{-1}$ and conductivity of pure water is 6×10^{-8} scm^{-1}

Soln: $K_{BaSO_4} = K_{total} - K_{water}$

$$= (3.06 \times 10^{-6} - 0.06 \times 10^{-6})\ scm^{-1}$$

$$= 3 \times 10^{-6}\ scm^{-1}$$

$$\Lambda^\circ_{eq,\,BaSO_4} = \Lambda_{eq,\,BaSO_4} = \frac{1000 \times K}{\text{Normality}}$$

$$150 = \frac{10^3 \times 3 \times 10^{-6}}{N}$$

$$N = \frac{3 \times 10^{-3}}{1.5 \times 10^{2}} = 2 \times 10^{-5}$$

Molarity = Solubility (s) = $\frac{N}{2} = 10^{-5}$ M

$$\boxed{K_{sp} = s^2 = 10^{-10}}$$ Ans

Q. N.Qmf IIT-2006

We have a saturated solution of AgBr, K_{sp} of AgBr is 12×10^{-14}. If 10^{-7} mol of $AgNO_3$ are added to 1L of this solution, find conductivity of this solution in Sm^{-1}.

Given Λ°, $Ag^+ = 6 \times 10^{-3}$ $Sm^2 mol^{-1}$

, $Br^- = 8 \times 10^{-3}$ $Sm^2 mol^{-1}$

, $NO_3^- = 7 \times 10^{-3}$ $Sm^2 mol^{-1}$

Sol^n :

$$AgBr(s) \rightleftharpoons Ag^+(aq) + Br^-(aq)$$

$$s \qquad (s + 10^{-7}) \qquad s$$

$\boxed{NO_3^- = 10^{-7} M}$

$$12 \times 10^{-14} = (s + 10^{-7}) \times s$$

$$s^2 + 10^{-7} s - 12 \times 10^{-14} = 0$$

$$s = \frac{-10^{-7} + \sqrt{10^{-14} + 48 \times 10^{-14}}}{2}$$

$$= 3 \times 10^{-7} M$$

$[Ag^+] = 4 \times 10^{-7}$ M

$[Br^-] = 3 \times 10^{-7}$ M

$[NO_3^-] =$ ~~10^{-7}~~ 10^{-7} M.

$$\lambda_{Ag^+} = \frac{K_{Ag^+}}{1000 \times 4 \times 10^{-7}} = 6 \times 10$$

$$\Rightarrow K_{Ag^+} = 24 \times 10^{-7} \text{ Sm}^{-1}$$

Sly

$$K_{Br^-} = 24 \times 10^{-7} \text{ Sm}^{-1}$$

$$K_{NO_3^-} = 7 \times 10^{-7} \text{ Sm}^{-1}$$

$$K_T = K_{Ag^+} + K_{Br^-} + K_{NO_3^-}$$

$$= 55 \times 10^{-7} \text{ Sm}^{-1}$$ Ans.

(*) Conductometric Titration–

→ It is based on the concept that during titration process, one ion is replaced by other and conductance of solution changes.

→ Note that $H^+(aq)$ and $OH^-(aq)$ ions have exceptionally high ionic conductivity, which is explained by Grotthus mechanism (Proton transfer)

$$\lambda_{H^+} \approx 7 \times \lambda^\circ_{Na^+}$$

→ H^+ and OH^- ions are considered to be special ions in Conductometric titration.

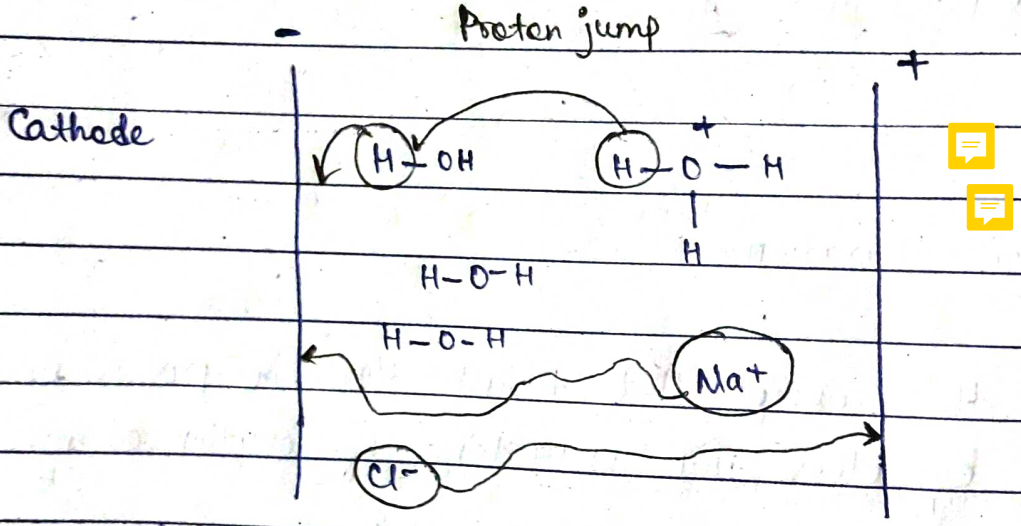
$$\lambda^\circ_{H^+} > \lambda^\circ_{OH^-}$$

$\rightarrow \lambda_{Li^+} < \lambda_{Na^+} < \lambda_{K^+} < \lambda_{Rb^+} < \lambda_{Cs^+}$

→ Greater is the extent of hydration, (charge same) greater will be the hydrated size, lower will be the ionic mobility and lower λ value.

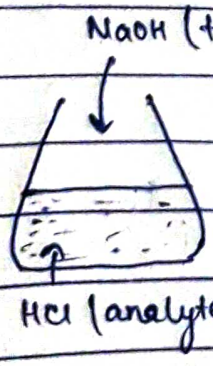
→ $\lambda_{Mg^{2+}} > \lambda_{Na^+}$

Proton jump in case of H^+ and OH^- :



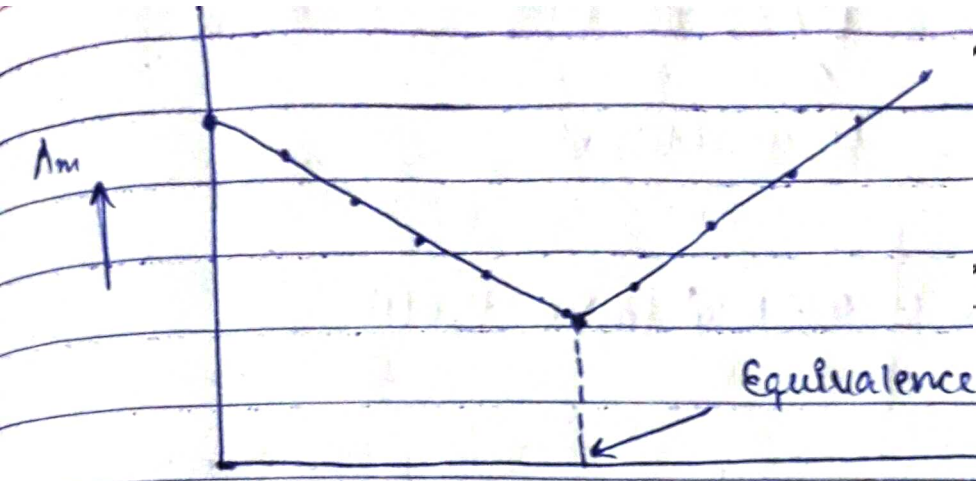
eg Titration of SA V/s SB :

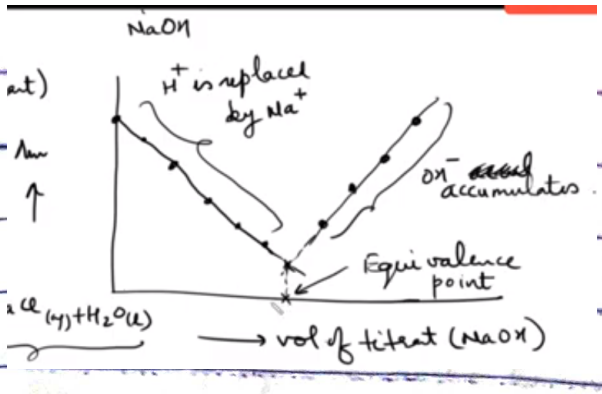
↑ HCl ↑ NaOH



titrant (taken drop by drop)

$HCl_{(aq)} + NaOH_{(aq)} \longrightarrow NaCl_{(aq)} + H_2O_{(l)}$





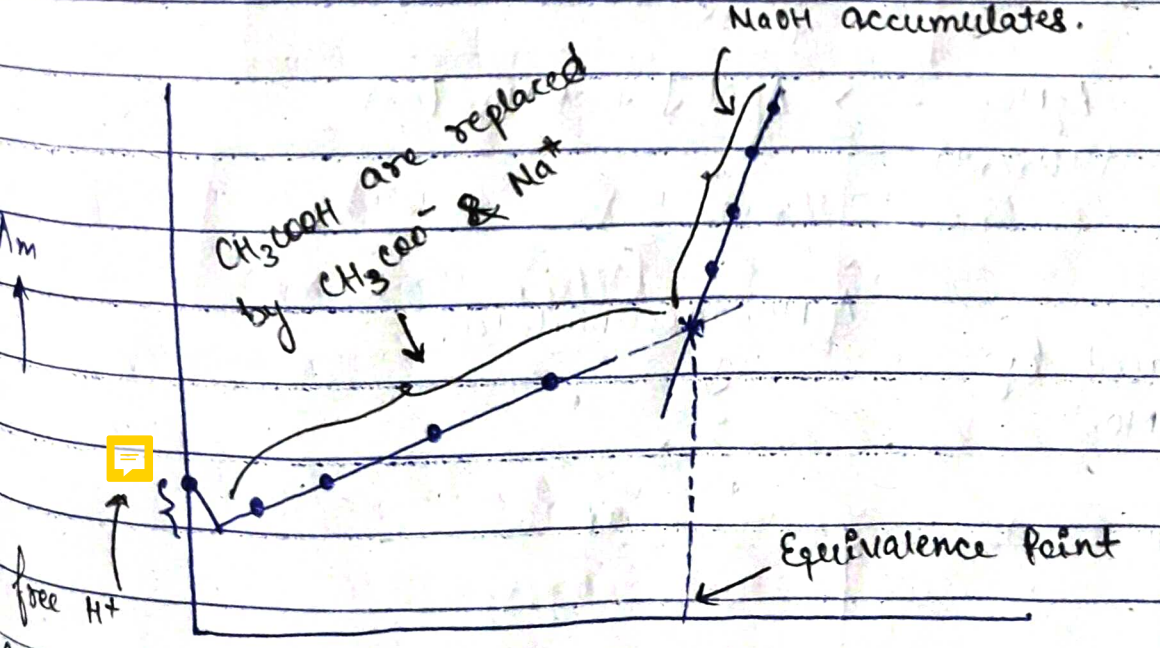
→ Volume of titrant (NaOH)

$N_1V_1 = N_2V_2$

eg. Titration of WA Vs SB

(CH_3COOH) ($NaOH$)

$CH_3COOH(aq) + NaOH \longrightarrow CH_3COO^-Na^+ + H_2O(l)$



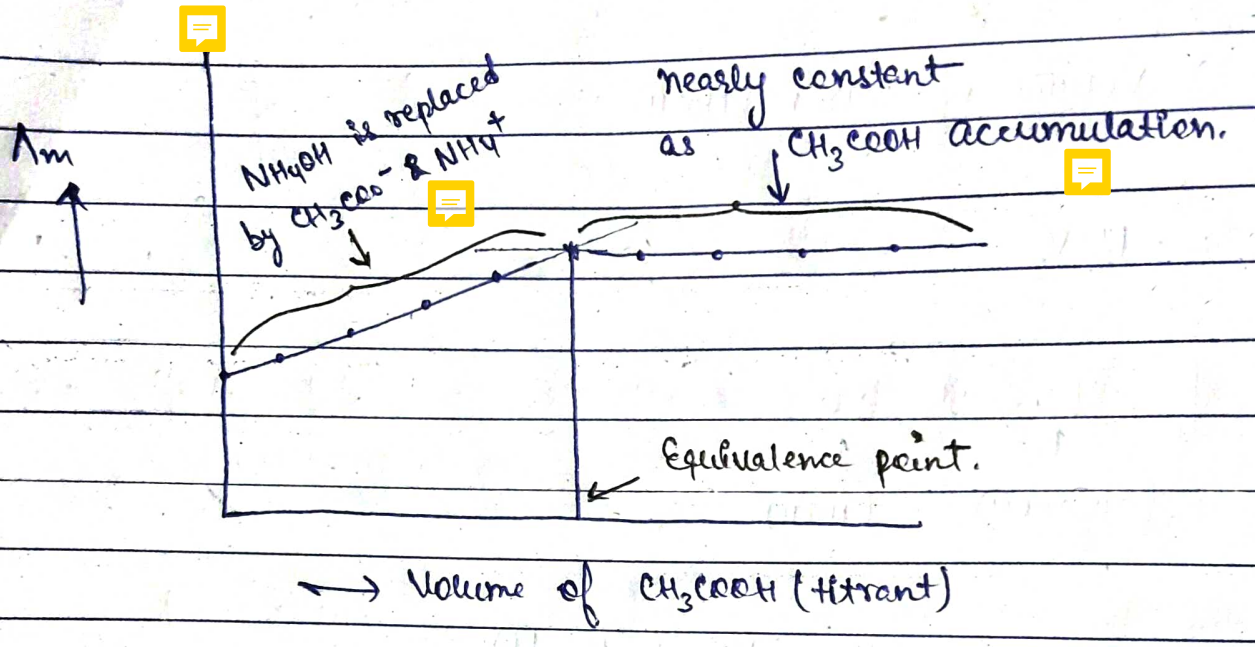
→ Volume of NaOH

eg. Titration of WB V/s WA

NH_4OH ← WB

CH_3COOH (titrant) ← WA

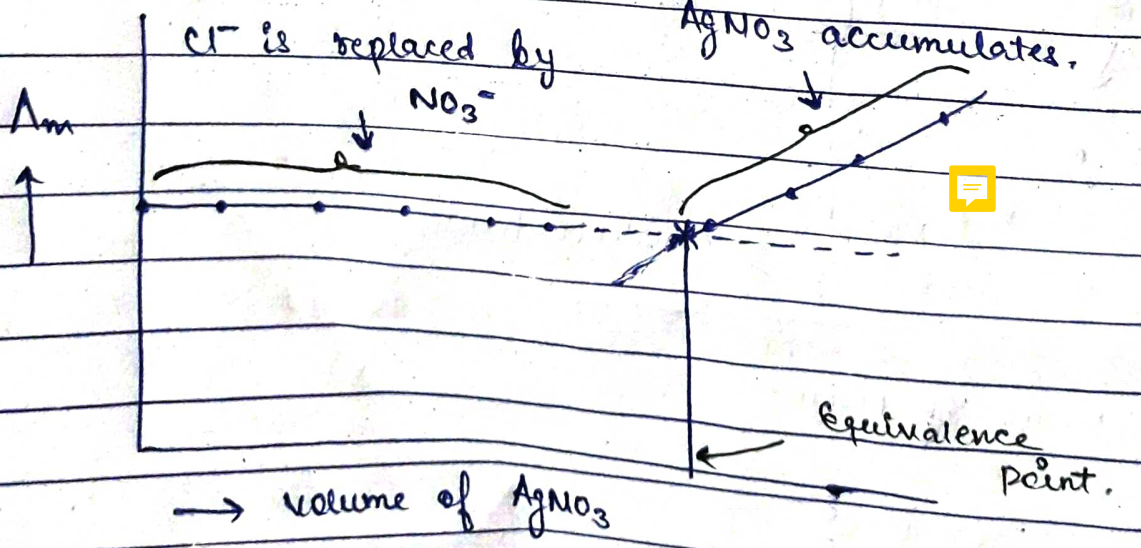
$$NH_4OH + CH_3COOH \longrightarrow CH_3COO^-NH_4^+(aq) + H_2O(l)$$



IIT

Q. Draw Λ_m vs vol. of $AgNO_3$ added

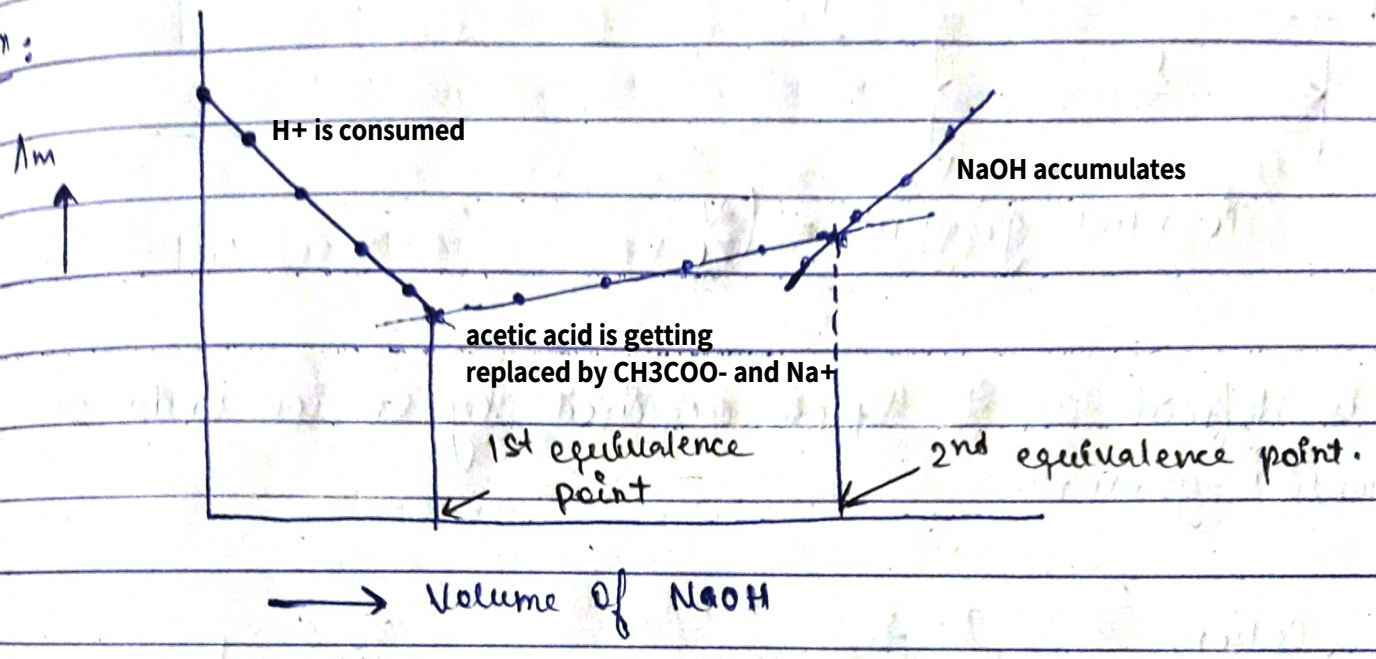
$$KCl(aq) + \underset{\text{(titrant)}}{AgNO_3} \longrightarrow AgCl\downarrow + KNO_3(aq)$$



Draw Λ_m v/s vol. of NaOH graph

$(HCl + CH_3COOH)$ + NaOH (titrant)

Analyte



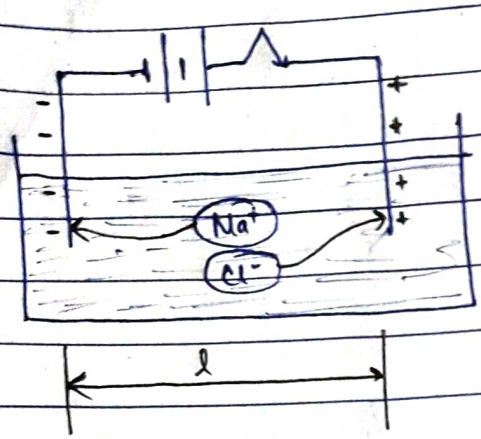
$HCl + NaOH \longrightarrow NaCl + H_2O$ (1st equivalence point)

$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ (2nd equivalence point)

numerical

(Never seen any Question in IIT on it)

(*) Ionic Mobility – (u_+ or u_-)



Potential gradient = $\left(\frac{V}{l}\right)$

→ It is defined as the speed acquired by an ion under unit potential gradient.

For Cation,

ionic mobility, $$u_+ = \frac{V_+}{(V/d)} = \frac{\lambda_+}{Z_+ \times F}$$

Similarly, For Anion,

ionic mobility, $$u_- = \frac{V_-}{(V/d)} = \frac{\lambda_-}{Z_- \times F}$$

V_+ → speed of cation

V → Potential difference applied

l → Separation between electrodes.

λ_+ → ionic conductance or ionic conductivity of cation, $S cm^2 mol^{-1}$

λ_- → ionic conductance or of anion.

Z_+ → ~~charge~~ unit of charge on cation

eg Mg^{2+}, $Z_+ = 2$

eg. Al^{3+}, $Z_+ = 3$

z_- = unit of charge on anion (magnitude)

e.g Cl^-, $z_- = 1$

SO_4^{2-}, $z_- = 2$

→ Absolute ionic mobility (u°_+ or u°_-)

↳ ionic mobility at ∞ dilution.

$$u^\circ_+ = \frac{v^\circ_+}{(V/d)} = \frac{\lambda^\circ_+}{z_+ \times F}$$ ← limiting ionic conductivity.

$$u^\circ_- = \frac{v^\circ_+}{(V/d)} = \frac{\lambda^\circ_-}{z_- \times F}$$

$F = 96,500\ C$

Unit : $\frac{cm/sec}{V/cm} = cm^2 V^{-1} sec^{-1}$ in CGS

$= m^2 V^{-1} sec^{-1}$ in SI

* Transport Number (t_c or t_a) or (t_+ or t_-)

Transport number is defined as the fraction of current carried by a constituent ion of an electrolyte.

$$t_+ = \frac{u_+}{u_+ + u_-}, \quad t_- = \frac{u_-}{u_+ + u_-}$$

↑ dimensionless No.

Q. A potential difference of 4.9 volt is applied across two electrodes placed 9.8 cm apart. How far would NH_4^+ be expected to move in 1 hr in a dilute sol^n of ammonium salt at 25°C. Ionic conductance of NH_4^+ at 25°C
$= 96.5\ scm^2 mol^{-1}$

Also calculate ionic mobility of NH_4^+ ion.

Sol^n $u_{NH_4^+} = \frac{V_+}{(V/d)} = \frac{\lambda_+}{z_+ \times F} = \frac{96.5}{1 \times 96500} = 10^{-3}\ cm^2 V^{-1} sec^{-1}$

$\frac{V^+}{(4.9/9.8)} = 10^{-3} \Rightarrow V_+ = 0.5 \times 10^{-3}$ cm/sec

$'d' = 0.5 \times 10^{-3} \times 3600$
$= 1.8$ cm Ans.

(*) Cells and Batteries : (Read NCERT)

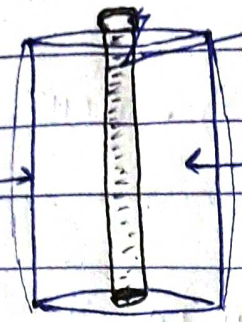
→ Cell refers to a single galvanic arrangement whereas battery is when two or more cells are connected in series to get higher voltage.

Types → Primary cell (Non-Rechargeable type)
eg. dry cell, Hg-cell

→ Secondary cell (Rechargeable type)
eg. Ni-cd cell, Lead storage Battery.

* Primary cell : (Scientist)

① Dry cell : (Leclanche cell)



Anode Reaction : $Zn(s) \longrightarrow Zn^{2+} + 2e^-$

Cathode : $2MnO_2 + 2NH_4^+ + 2e^- \longrightarrow Mn_2O_3(s) + 2NH_3 + H_2O$
(or MnO·OH)

$Zn^{2+} + NH_3(aq) \longrightarrow [Zn(NH_3)_4]^{2+}_{(aq)}$

→ Cell potential is 1.5 volt and it decreases over use.

→ Even when it is not in use, NH_4Cl corrodes Zn container and leaks out after a long time.

② Hg - cell (or Mercury cell or Button cell)

↳ used in wrist watch, Calculator etc.

Anode : Zn-Hg amalgam

Cathode : Paste of HgO + Carbon.

Electrolyte : KOH + ZnO

Anode Rkn : $\underset{\text{(amalgam)}}{Zn} + 2OH^- \longrightarrow ZnO(s) + H_2O + 2e^-$

Cathode Rkn : $HgO(s) + H_2O + 2e^- \longrightarrow Hg(l) + 2OH^-$

Cell Rkn. : $\underset{\text{(Amalgam)}}{Zn(s)} + HgO(s) \rightarrow ZnO(s) + Hg(l)$

Cell potential ≈ 1.35 volt

* Cell potential remains constant over its use.

* It is due to that all active masses involved in cell rkn. are constant.

(*) Rechargeable Battery (Secondary battery):

(1) Ni-Cd cell : used in electronics devices

Anode : Cd. Cathode : $Ni(OH)_3$

Electrolyte : NaOH or KOH soln.

Anode rkn. $Cd(s) + 2OH^-(aq) \longrightarrow Cd(OH)_2(s) + 2e^-$

Cathode rkn. $2Ni(OH)_3 + 2e^- \longrightarrow 2Ni(OH)_2 + 2OH^-$

Cell rkn. $Cd(s) + 2Ni(OH)_3 \longrightarrow Cd(OH)_2 + 2Ni(OH)_2$

cell potential remains constant, just like in Hg cell

imp

(2) Lead Storage Battery :
(or lead acid Accumulator)

→ used in automobiles, power back-up etc.

Anode : lead (Pb)
Cathode : Lead coated with PbO_2
Electrolyte : aq. H_2SO_4 ($\simeq$ 38% by mass)

Anode rkn. $Pb(s) + SO_4^{2-}(aq) \longrightarrow PbSO_4(s) + 2e^-$

Cathode rkn. $PbO_2(s) + SO_4^{2-}(aq) + 4H^+ + 2e^- \longrightarrow PbSO_4(s) + 2H_2O(l)$

Cell Rkn. $Pb(s) + PbO_2(s) + \underbrace{2SO_4^{2-}(aq) + 4H^+}_{2H_2SO_4(aq)} \underset{\text{Recharging}}{\overset{\text{discharging}}{\rightleftharpoons}} 2PbSO_4(s) + 2H_2O(l)$

V. imp

While discharging lead storage battery, 1 'F' of electricity is withdrawn from battery, 1 mol of H_2SO_4 is consumed.

i.e. valency factor of H2SO4 in this reaction is 1

Q. What current for 1 hr can be withdrawn from lead storage battery if 49 g of H_2SO_4 is consumed.

Soln No. of 'F' = no. of mol H_2SO_4 consumed

$$\frac{i \times t}{96500} = \frac{49}{98} = 0.5$$

$$i = \frac{0.5 \times 96500}{3600} = \frac{48}{36} = \frac{4}{3} \text{ amp.}$$

IIT- Adv'

Q. During discharging of lead Storage battery,

H_2SO_4 (3L, 40% W/W, d = 1.4 g/ml)

$\longrightarrow$ H_2SO_4 (3L, 20% W/W, d = 1.2 g/ml)

Calculate no. of amp-hrs derived from battery.

Sol^n : No. of moles of H_2SO_4 consumed

$$\text{No. of 'F'} = \frac{(3000 \times 1.4 \times 0.4 - 3000 \times 1.2 \times 0.2)}{98}$$

$$\frac{i \times t}{96500} = \frac{3000(0.56 - 0.24)}{98}$$

$$i \times t = \frac{3000 \times 0.32}{98} \times 96500 \text{ amp-sec}$$

$$\text{No. of amp-hrs} = \frac{3000 \times 0.32 \times 96500}{98 \times 3600} \text{ amp-hrs. Ans.}$$

* FUEL CELL :

→ used in Rockets

→ can be used in public transport / cars in future.

→ costly (a disadvantage)

→ Most common and successful fuel cell is H_2/O_2 fuel cell (→ pollution free)

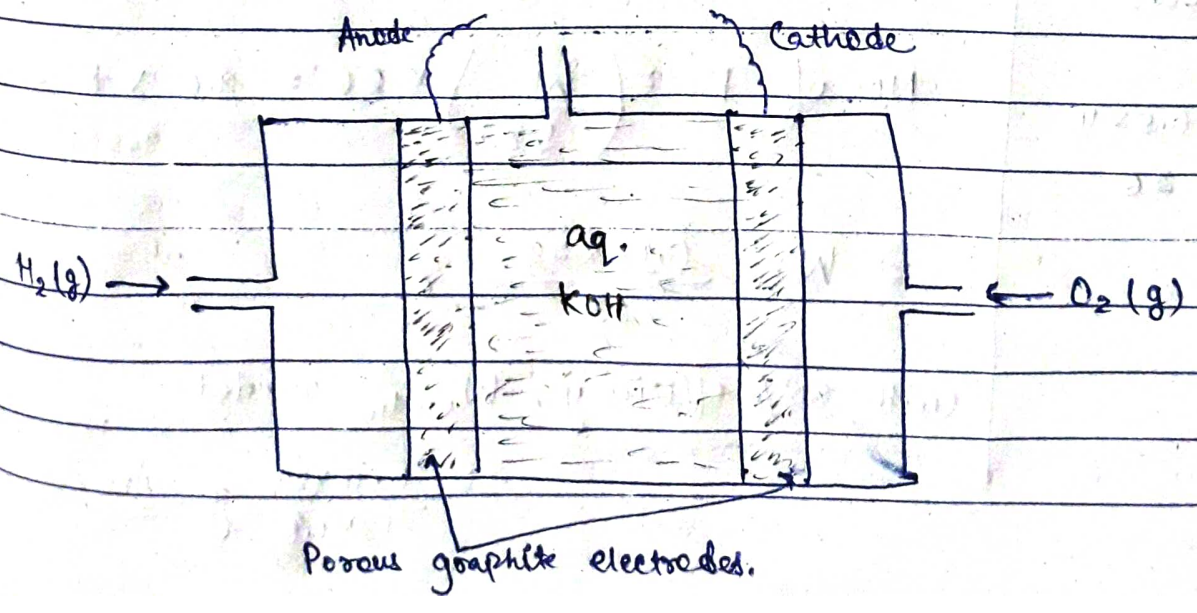
$$H_2 + \frac{1}{2}O_2 \longrightarrow H_2O(l)$$, Cell reaction.

↑ end product is non-polluting.

→ fuel cell are much more efficient (≈ 70% efficiency) in comparison to thermal power plants (≈ 40%)

→ fuel (H_2) and oxidizer (O_2) are continuously supplied in fuel cell and hence we can obtain electrical energy continuously.

eg. H_2/O_2 fuel cell :



Anode Rkn. $H_2 + 2OH^- \longrightarrow 2H_2O + 2e^-$

Cathode Rkn. $\frac{1}{2}O_2 + H_2O + 2e^- \longrightarrow 2OH^-(aq)$

Cell Rkn. $H_2 + \frac{1}{2}O_2 + H_2O \longrightarrow 2H_2O$

OR

$H_2 + \frac{1}{2}O_2 \longrightarrow H_2O$

Non-polluting and product.

eg. of fuel cell : CH_4/O_2, C_4H_{10}/O_2 etc.

AIIMS

Q. What mass of butane is needed in a butane-O_2 fuel cell to obtain a current of 9.65 amp. for 30 minutes with current efficiency of 80%.

Soln :

$C_4H_{10} + 6.5O_2 \longrightarrow 4CO_2 + 5H_2O$

$4x + 10 = 0$

$x = -2.5$

(+4)

$Vf = 6.5 \times 4$

$= 26$

No. of 'F' $= \left(\frac{W}{58}\right) \times 26 = \frac{9.65 \times 30 \times 60}{96500}$

$\Rightarrow W = \frac{0.18 \times 29}{13}$ g

With 80% efficiency, $W_{C_4H_{10}}$ needed

$= \frac{0.18 \times 29}{13 \times 0.8}$ g Ans.

(*) Corrosion :

A slow coating of oxides / sulphides / carbonates formation on surface metal in atmospheric air.

Corrosion of iron in moist air is called 'Rusting'

eg. Rusting of iron ($Fe_2O_3 \cdot xH_2O$)

eg. Tarnishing of Silver (H_2S (a pollutant in air) attacks Ag metal in presence of O_2 and form a coating of Ag_2S)

eg. Copper / Bronze / Brass – a green coating forms (due to formation of basic Copper Carbonate) $[Cu(OH)_2 \cdot CuCO_3]$

* Rusting of Iron :

→ A porous coating of iron oxide (i.e $Fe_2O_3 \cdot xH_2O$) forms on iron surface in moist air.

→ Anode : $Fe(s) \longrightarrow Fe^{2+} + 2e^-$, $E^\circ_{oxid} = 0.44\ V$

Cathode : $\frac{1}{2}O_2 + 2H^+(aq) + 2e^- \longrightarrow H_2O(l)$, $E^\circ_{Red} = 1.23\ V$

Electrochemical cell : $Fe(s) + \frac{1}{2}O_2(g) + 2H^+(aq) \longrightarrow Fe^{2+}(aq) + H_2O(l)$,

$E^\circ_{cell} = 1.67$ Volt

H^+ are due to CO_2 (Pollutant) being dissolved in H_2O forming H_2CO_3 which provides required H^+.

→ $Fe^{2+} + O_2 \longrightarrow Fe_2O_3 \cdot xH_2O$
$+ H_2O$ (moisture) (Reddish Brown)
(Porous Coating)

→ Due to porous coating of rust, a further attack of moist air occur on iron metal which decreases Strength of metal.

→ In case of Al/Zn, Ni/Cr, oxide coating is non-porous & protects the metal from further attack of air.

→ Rusting is very fast and severe problem in Saline water (eg. Sea water) as Salts in water make electrochemical process fast as these salts act as conductor.

→ Methods to prevent/minimize Rusting of iron:

- Paint or coating of bisphenol keeps the metal away from atmospheric air attack.
- Galvanization (i.e a coating of Zn metal on iron pipes/sheets)
- Cathodic protection (or Using a Sacrifical Anode)

→ A big iron object (eg. hulls of ships) is strapped with a more reactive metal (eg Mg / Zn)

→ Mg / Zn is preffered to have oxidation (anode) and iron object becomes cathode.